Class 9
Physics
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1. WORK, POWER AND ENERGY

WORKSHEET – 1

LEVEL – I

1) A	2) B	3) A	4) D	5) D
6) A	7) A	8) D	9) A	10) C
11) B	12) D	13) C	14) A	15) B
16) B	17) C			

LEVEL – II

1) C	2) B	3) B	4) C	5) C
6) C	7) B	8) B	9) C	10) D
11) B	12) B	13) C	14) A	15) B
16) A	17) B	18) C	19) A	20) B
21) D				

LEVEL – III

22) C	23) A, C, D	24) D	25) A	26) C
27) D	28) a-1; b-2; c-3; d-5		29) 60	30) A, B, D
31) A, B, C, D	32) A, B	33) A, B	34) D	35) B

SOLUTIONS

LEVEL – II

1. $W = (mg\sin\theta)s$ 2 W = Mgh
2. W = Mgh
3. W = $FS\cos\theta$
4. $W = \frac{MgL}{2}$
5. $W = \vec{F} \cdot \vec{S}$
6. conceptual
7. W = Area of F – S graph
8. $W = \frac{MgL}{2n^2}$
9. $W = FS, W_f = -$ fs

   F = f = μmg
11. Here, $\vec{S} = \vec{r}_2 - \vec{r}_1 = (\hat{i} - \hat{j} - 10\hat{k})$m

    $\vec{F} = (6\hat{i} - 2\hat{j} + \hat{k})N \Rightarrow W = \vec{F} \cdot \vec{S} = 6 + 2 - 10 = -2J$
12. $W = \frac{mgl}{2n^2}$
13. $W = \int_0^4 F dx$
14. $W_1 = \left(\frac{3}{2}\right) ab$
15. $W = \int_{x_1}^{x_2} \vec{F} \cdot \overrightarrow{dx}$

16. $W = \frac{1}{2}mv^2$

17. $W = \left(Mgh + \frac{1}{2}mu^2\right) - \frac{1}{2}mv^2$ 18. $W = F\cos\theta$ 19. $W = U_f - U_i$

20. Use the work energy theoram $W = \Delta K($ where $V = \frac{dx}{dt}\Big)$;

$W = \frac{1}{2}mv^2 - \frac{1}{2}mv^2$ t = 0, V = 6 m/s, t = 6sec, v = 6 m/s, W = zero

21. $W = \int \vec{F} \cdot \overrightarrow{d_r} = K\int_{r_A}^{r_B} \left[\frac{x\hat{\imath}}{(x^2+y^2)^{3/2}} + \frac{y\hat{\jmath}}{(x^2+y^2)^{3/2}}\right](dx\hat{\imath} + dy\hat{\jmath})$

$$= K\int_{r_A}^{r_B} \frac{1}{(x^2+y^2)^{3/2}}\left(d\left(\frac{x^2}{2}\right) + d\left(\frac{y^2}{2}\right)\right)$$

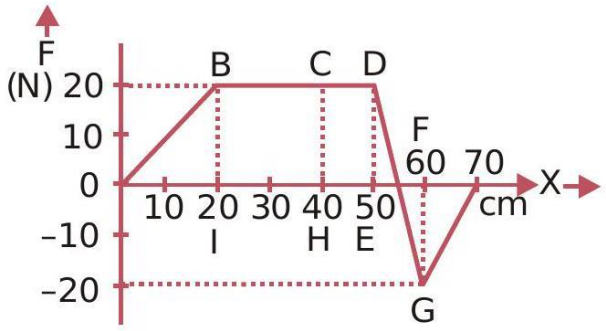
$$= K\int_{r_A}^{r_B} \frac{1}{(x^2+y^2)^{3/2}}(x^2+y^2) = K\int_{r_A}^{r_B} \frac{1}{2r^3}2(x^2)$$

$$= K\int_{r_A}^{r_B} \frac{2rdr}{2r^3} = K\int_{r_A}^{r_B} \frac{dr}{r^2} = K\left[-\frac{1}{r}\right]_{r_A}^{r_B}$$

$$= K\left[\frac{1}{r_A} - \frac{1}{r_B}\right] = K\left[\frac{1}{a} - \frac{1}{a}\right] = K \times 0\ W = 0$$

LEVEL – III

25. Work done at the end of displacement 20 cm



$\Rightarrow$ Area OBI $= \frac{1}{2} \times (20 \times 10^{-2}) \times 20 = 2$ joule

26. Work done at the end of displacement 40 cm. = Area of the figure OBCHIO = Area of OBI + Area of rectangle BCHI $= 2 + 20 \times 10^{-2} \times 20 = 6$ joule

$\Rightarrow$ Area OBI $= \frac{1}{2} \times (20 \times 10^{-2}) \times 20 = 2$ joule

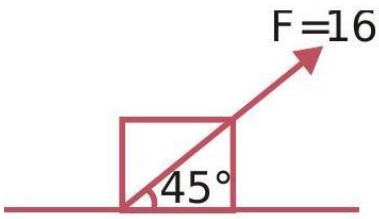
27. Work done at the end of displacement 60 cm :

= Area of triangle OBI + area of trapezium BDEI - area of $_{\Delta}$EGF

29. W = F.S = mg × s = 2 × 10 × 3 = 60 joule

30. a) work done by the applied force = W = FScos θ

$$= 16 \times 2.2 \times \cos 45^\circ = 16 \times 2.2 \times \frac{1}{\sqrt{2}} = 24.9 \text{ J}$$



b) work done by the normal force W = F × S × cos 90° = 0 J

c) W = F × S × cos 90° = 0 J

d) Total workdone = resultant force in the direction of the displacement × displacement.

$\therefore$ Total work done $W = 16 \times 2.2 \times \cos 45^\circ = 24.9$ J.

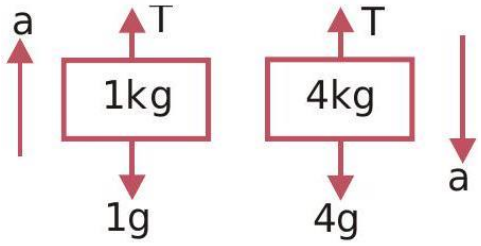
32 1) work done by gravity $W = (mg\sin\theta) \times 2$

$= 2 \times 10 \times \frac{\sqrt{3}}{2} \times 2 = 20\sqrt{3} = 34.64$ J

2) work done by a force of friction

$W = -\mu mg\cos\theta \times 2 = -\frac{1}{2} \times 2 \times 2 \times 10 \times \frac{1}{2} = -10$ J

33.



$T - g = a - \cdots - C^- - -(1); 4g - T = 4a$

Adding (1) and (2), we have

$3g = 5a \Rightarrow a = \frac{3g}{5}$.

From (1) $T = a + g = \frac{3g}{5} + g = \frac{8g}{5}$

1. The work done by gravity on 4 kg block in 2sec is $W = FS\cos\theta$

Here $F = 4 \times g = 40$ N. $s = \frac{1}{2}at^2 = \frac{1}{2} \times \frac{3}{5} \times 10 \times 4 = 12$ m; $\theta = 0^\circ$.

$\therefore W = 40 \times 12 \times \cos 0^\circ = 480$ N

2. The work done by string on 1 kg block in 2sec is $W = FS\cos\theta$

Here $F = T = \frac{8g}{5} = 16$ N,

$$s = \frac{1}{2} \times at^2 = \frac{1}{2} \times 6 \times 4 = 12 \text{ m} \quad \therefore \theta = 0^\circ \quad \therefore W = 16 \times 12 \times \cos 0^\circ = 192 \text{ J}$$

34. $W = mgh\left(1 - \frac{\rho_w}{\rho_b}\right) = 40 \times 10 \times 12\left(1 - \frac{1000}{3000}\right) = 3200$ J

35. $W = \int_{(0,0)}^{(1,1)} \vec{F} \cdot \overrightarrow{ds}$; $W = \int_{(0,0)}^{(1,1)} (y\hat{\imath} + x^2\hat{\jmath}) \cdot dx\hat{\imath} + dy\hat{\jmath} + dz\hat{k}$

$$W = \int_{(0,0)}^{(1,1)} x^2dy + ydx; \qquad W = \int_{(0,0)}^{(1,1)} y^2dy + xdx$$

$$W = \left(\frac{y^3}{3}\right)_{(0,0)}^{(1,1)} + \left(\frac{x^2}{2}\right)_{(0,0)}^{(1,1)}; \quad W = \frac{5}{6}J$$

WORKSHEET – 2

LEVEL – I

1) C	2) D	3) A	4) D	5) A
6) D	7) C	8) C	9) B	10) A

LEVEL – II

1) C	2) C	3) B	4) C	5) B
6) B	7) C	8) C	9) B	10) B
11) D	12) C	13) D	14) D	15) D

16) C 17) C 18) C

LEVEL – III

19) A, C 20) A 21) A 22) C 23) A

24) a-1; b-2; c-3; d-4 25) 45 26) A, C, D 27) D

28) A, C 29) C 30) a-q,s; b-p,s; c-p,s; d-q,s

SOLUTIONS

LEVEL – II

1. $U = \frac{1}{2}Kx^2 \Rightarrow 1 = \frac{1}{2} \times K \times (20 \times 10^{-2})^2 \Rightarrow K = 50 Nm^{-1}$
2. Work done $w = \frac{1}{2}kx$
3. $\frac{1}{k} = \frac{1}{k_1} + \frac{1}{k_2}$
4. $K.E = \frac{1}{2}Kx^2, F = Ky$
5. $E = \frac{1}{2}Kx^2$
6. $P.E = \frac{1}{2}kx^2 = \frac{1}{2}k\left[\frac{F}{k}\right]^2 \ (\because F = kx)$

   $P.E = \frac{1}{2}\frac{F^2}{k} \ \left(\because x = \frac{F}{k}\right)$

   $\therefore \frac{U_1}{U_2} = \frac{k_2}{k_1} = \frac{200}{100} = 2:1$
7. $E = \frac{1}{2}Kx^2$

   $\Rightarrow \frac{1}{2}kx^2 = 1$

   $\Rightarrow k = \frac{2}{x^2} = \frac{2}{(40 \times 10^{-2})} = 12.5 Nm^{-1}$
8. As $\frac{1}{2}\ K_1x_1^2 = \frac{1}{2}\ K_2x_2^2 \ \therefore \frac{x_1}{x_2} = \sqrt{\frac{K_2}{K_1}}; \frac{F_1}{F_2} = \frac{K_1x_1}{K_2x_2} = \frac{K_1}{K_2}\sqrt{\frac{K_2}{K_1}} = \sqrt{\frac{K_1}{K_2}}$
9. $P.E = \frac{1}{2}Kxx^2$
10. $\frac{1}{2}kx^2 = \frac{1}{2}mv^2; 800 \times \left(\frac{5}{100}\right)^2 = 2 \times v^2$

    $\Rightarrow \frac{800 \times 25}{10000} = 2v^2$

    $\Rightarrow 2 = 2v^2; v^2 = 1; v = 1 \ m/s$
12. $W = \frac{1}{2}k(x_2^2 - x_1^2)$

    $W = \frac{1}{2} \times 5 \times 10^3((0.1)^2 - (0.05)^2)$

    $W = \frac{1}{2} \times 5 \times 10^3 \times 0.75 \times 10^{-2}$

    $W = \frac{1}{2} \times 5 \times 0.75 \times 10; W = 25 \times 0.75; W = 18.75 \ J$
13. $t = \sqrt{\frac{2\ h}{g}} = \sqrt{\frac{2 \times 5}{10}} = 1 sec,$

    distance travelled by lift in 1sec is 5 m,

    Total distance travelled by the bolt is

    $5 + 5 = 10$ m.

    heat produced is work done

$= mgh$

$= 0.5 \times 10 \times 10 = 50j$

14,15,16. $U = \frac{1}{2}Kx^2$

17. $\frac{1}{4} = \frac{1}{k} + \frac{1}{k} + \frac{1}{k} + \frac{1}{k}$

$\frac{1}{4} = \frac{4}{k}$

$k = 16$

parallel, $k_{eff} = 4k = 4 \times 16 = 64$

18. Natural length $I_1 = AB = \frac{2\pi r}{4}$; Extended length $I_2 = AC = 3 \times \frac{2\pi r}{4}$

Increases in length $x = I_2 - I_1 = \frac{3}{2}\pi r - \frac{\pi r}{2} = \frac{3\pi r - \pi r}{2} = \pi r$

Energy stored $= \frac{1}{2}Kx^2 = \frac{1}{2}K\pi^2 r^2$

LEVEL – III

22. $\because$ Work done $w = \frac{1}{2}k(x_2^2 - x_1^2) = \frac{1}{2} \times 8 \times 10^2[(0.15)^2 - (0.05)^2] = 8$ J

23. Since $K_1 > K_2$, Therefore, P will develop greater restoring force than Q. So, applied force on P will be more than the applied force on Q. Extension is same in both the cases. So, more work will be done on P.

24. Potential energy stored in the spring when it is extending by x is $U_1 = \frac{1}{2}kx^2$

Potential energy stored in the spring when it is further extended by x is $U_2 = \frac{1}{2}(x + x)^2 = 2kxx^2$

$\therefore$ work done = gain in potential energy

$$= U_2 - U_1 = 2kx^2 - \frac{1}{2}kx^2 = \frac{3}{2}x^2$$

28. $W = 4$ J, $x = 10$ cm $= 10^{-1}$ m, $k = ?$

$$W = \frac{1}{2}kxx^2 \Rightarrow 4 = \frac{1}{2}k(10^{-1})^2$$

$$\Rightarrow k = \frac{8}{10^{-2}} = 800 \text{ N/m}$$

Total stretching $x^1 = 10 + 10 = 20$ cm $= 20 \times 10^{-2}$ cm $= 0.2$ m

Total work done $= \frac{1}{2}kx'^2; W^1 = \frac{1}{2} \times 800(0.2)^2 = 16$ J

$\therefore$ Extra work required $= W^1 - W = 16 - 4 = 12$ J

Total stretching $x^1 = 10 + 10 = 20$ cm $= 20 \times 10^{-2}$ cm $= 0.2$ m

Total work done $= \frac{1}{2}kx'^2; W^1 = \frac{1}{2} \times 800(0.2)^2 = 16$ J

Extra work required $= W^1 - W = 16 - 4 = 12$ J

29. Let m be the mass of the block.

Initial elongation of the spring will be

$x_1 = \frac{mg}{k}$

When the force F is applied, work done by F and gravity is used to increase the elastic potential energy of spring.

Hence $(F + mg)x_0 = \frac{1}{2}k(x_i + x_0)^2 - \frac{1}{2}kx_i^2$

From equation (1) and (2),

we get $x_0 = \frac{2F}{k}$. Work done by applied force $F = Fx_0$

30. Conceptual

WORKSHEET – 3

LEVEL – I

1) C	2) A	3) D	4) C	5) A
6) B	7) C	8) B	9) C	10) D
11) A	12) C	13) B	14) A	15) D

LEVEL – II

1) 1	2) B	3) D	4) B	5) B
6) B	7) C	8) A	9) D	10) A
11) C	12) B	13) D	14) A	15) C
16) C	17) B	18) C	19) A	20) D
21) B				

LEVEL – III

22) A, B, C	23) C	24) C	25) C	
26) B	27) 8	28) a-4; b-1; c-2; d-3		29) A, B
30) a-r; b-t; c-q; d-p,s		31) B	32) C	33) D

SOLUTIONS

LEVEL – II

1. $\text{K.E} \propto \text{V}^2; \frac{\text{KE}_1}{\text{KE}_2} = \frac{\text{V}_1^2}{\text{V}_2^2} = \frac{\text{V}^2}{(2\ \text{V})^2} = \frac{1}{4}$
2. $\text{W} = \Delta\text{KE} = \frac{1}{2}\ \text{m}(\ \text{V}_2^2 - \text{V}_1^2)\ \text{W}_1 = \frac{1}{2}\ \text{m}(400 - 100)$
   $\text{W}_2 = \frac{1}{2}\ \text{m}(100 - 0)$
   (2) From (1) and (2),
   $\frac{\text{W}_1}{\text{W}_2} = \frac{300}{100} = 3; \text{W}_1 = 3\ \text{W}_2.$
3. K.E αt^2
4. $K \cdot E = \frac{1}{2}M(\vec{V} \cdot \vec{V})$
5. $K \cdot E = \frac{1}{2}MV^2, 4K = \frac{1}{2}M(2)^2$
6. $K.E = \frac{P^2}{2m}$
7. $\left(\frac{\Delta P}{P}\right)^2 = \frac{\Delta KE}{KE}$
8. $\text{KE} = \text{mgl}[1 - \cos\theta]$
9. $W = \frac{1}{2}mv^2$
10. $KE = F.S$
11. $m_1v_1 = m_2v_2, KE = \frac{1}{2}m_2v_2^2$
12. K.E αm
13. $KE = \frac{p^2}{2m}$
14. $\text{F.S} = \Delta KE$
15. K.E αt^2

16. $m_1v_1 = m_2v_2, KE = \frac{1}{2}m_2v_2^2$

17. $\text{F.S} = \Delta KE$

18. $\text{KE} = \frac{1}{2}mv^2$

19. $KE = \frac{1}{2}mv^2$ and $v = \frac{d}{t}$

20. $\frac{1}{2}Mu^2 = \frac{1}{2}\left(\frac{1}{2}mv^2\right), M = 2m$

21. By work-energy theorem

$$W = \left(mgh + \frac{1}{2}mv^2\right) - \frac{1}{2}mv^2$$

$$W = \left(\frac{10}{1000} \times 10 \times 50 + \frac{1}{2} \times \frac{10}{1000} \times (1000)^2\right) - \frac{1}{2} \times \frac{10}{1000} \times (500)^2$$

$$W = (5 + 5000) - 1250 = W = 3755 \text{ J}$$

LEVEL – III

24. $K \cdot E = \frac{P^2}{2m} = \frac{2^2}{2\times2} = 1J$

25. $K.E = \frac{P^2}{2m} \Rightarrow P^2 = 2m \times K.E = 2 \times 4 \times 2 = 16 \text{ kg.m/s}$

$\Rightarrow \text{P} = 4 \text{ kg} \cdot \text{m/s}$

26. $\text{K.E} = \frac{\text{P}^2}{2 \text{ m}} = \frac{12\times12}{2\times6} = \frac{144}{12} = 12 \text{ J}$

27. $K.E = \frac{P^2}{2m} \Rightarrow P^2 = 2 \text{ m} \times \text{K}.E = 2 \times 16 \times 2 = 64 \text{ kg} \cdot \text{m/s} \Rightarrow P = 8 \text{ kg} \cdot \text{m/s}$

28. $F_x = -6 \times N$

$$W = \int_{x=3m}^{x=4m} Fdx$$

$$W = \int_3^4 -6xdx$$

$$= -\frac{6}{2}[x^2]_3^4 = -3[16 - 9] = -3[7] = -21J$$

32. $\text{W}_{\text{net}} = \Delta\text{KE} = \text{K}_\text{f} - \text{K}_\text{i}$

$$-21 = \frac{1}{2} \times \text{m}(\text{v}^2 - \text{u}^2); \quad -21 = \frac{1}{2} \times 2(\text{v}^2 - 64)$$

$$\text{v}^2 = 64 - 21;$$

$$\text{v} \approx 6.6 \text{ m/s}$$

33. $\text{W}_{\text{net}} = \Delta\text{KE}; \Delta\text{KE} = \frac{1}{2} \times 2(25 - 64) = -39$

$W = \int F \cdot dx;$ $\quad W = -6\int_3^x xdx = -6\left(\frac{x^2}{2}\right)_3^x$

$W = -3(x^2 - 9);$ $\quad -3(x^2 - 9) = -39$

$x^2 - 9 = 13;$ $\quad x^2 = 22$

$x = \sqrt{22} \Rightarrow x \approx 4.7m$

WORKSHEET – 4

LEVEL – I

1) B	2) D	3) D	4) C	5) A
6) A	7) B	8) D	9) B	10) A
11) B	12) C			

LEVEL – II

1) A	2) B	3) B	4) C	5) D
6) B	7) C	8) C	9) A	10) A
11) B	12) B	13) D	14) B	15) C

LEVEL – III

16) A, C	17) A	18) B	19) B	20) A
21) a-1; b-2,3; c-3,4; d-1,3		22) 3	23) 1	24) A, B
25) A, C, D	26) A			

SOLUTIONS

LEVEL – II

1. $PE = (lbh)dg\frac{h}{2}$
2. $\frac{1}{2}mv^2 = \frac{1}{2}Kx^2$
3. $\frac{1}{2}mv^2 = \frac{1}{2}Kx^2$
4. $F = 0$ where $\frac{dU(x)}{dx} = 0 \Rightarrow a = 0$ 5. $U = \frac{1}{2}kx^2$
5. $P.E = mgh = 10 \times 9.8 \times 20$

   P.E = 1960J
6. $U = \frac{1}{4\pi\epsilon_0}\frac{q_1q_2}{r}$; $\frac{U_1}{U_2} = \frac{r_2}{r_1}$; 2:1
7. Increase in elastic P.E = Decrease in PE $\Rightarrow mgx = \frac{1}{2}kx^2 \Rightarrow x = \frac{2mg}{k}$

   $PE = \frac{1}{2}kx^2$
8. $W_N = N \times 2.2 \times \cos 90°$; $W_N = 0$
9. $kx = mg$; $W = \frac{1}{2}kxx^2$

   $W = \frac{1}{2}kx \cdot x$; $W = \frac{1}{2}mgx$

   $W = \frac{1}{2} \times 5 \times 10 \times 3$; $W = 75$ J
11. $U = -\int F dx = -\frac{1}{2}Kx^2$
12. $F(x) = -kx + ax^3; F_{(x)} = 0$; for $x = 0\ x = \sqrt{\frac{k}{a}}$; So slope is zero at $x = 0\ x = \sqrt{\frac{k}{a}}$
13. put $F = \frac{-du}{dx} = 0$, $x = \pm\sqrt{\frac{2a}{b}}$

    so u is max (because $\frac{d^2u}{dx^2}$ is -ve for x); so, it's unstable equilibrium

LEVEL – III

16. U is minimum if $\frac{dU}{dx} = 0$ and $\frac{d^2U}{dx^2} > 0$ Now $\frac{dU}{dx} = \frac{d}{dx}(ax^2 - bx) = 2ax - b$

    So, $\frac{dU}{dx} = 0$ at $x = \frac{b}{2a}$; $\frac{d^2U}{dx^2} = \frac{d}{dx}(2ax - b) = 2a$ which is positive

    For $x = \frac{b}{2a}$; $U_{\min} = a\left(\frac{b}{2a}\right)^2 - b\left(\frac{b}{2a}\right) = -\frac{b^2}{4a}$

Force acting on the system is $F = -\frac{dU}{dx} = b - 2ax$

18,19, & 20. Equilibrium occur at point A, C and E because F = 0 at these points.

$$\text{We know } \frac{d^2U}{dx^2} = \frac{-d}{dx}\left[-\frac{dU}{dx}\right] \left(\because F = \frac{-dU}{dx}\right)$$

$-\frac{dF}{dx} = -ve$ slope of $F - x$ graph. At point A slope of $F - x$ curve is tve

$\therefore \frac{d^2U}{dx^2}$ is negative, therefore potential energy is maximum at A and therefore point A is position of unstable equilibrium.

Similarly, we can say that E is the position of unstable equilibrium and C is the position of stable equilibrium position.

22. $U = \frac{1}{2}kx$;

$U_1 = \frac{1}{2}k \times 1^2$

$k = 2U$;

$U_2 = \frac{1}{2} \times 2U \times 4^2$

$U_2 = 16U$;

$U_2 - U_1 = 5 \times 3U$

23. Initial potential energy of body $= mg(h + x)$

final potential energy = spring potential energy $= 1/2Kx^2$

according to law of conservation of energy,

initial potential energy = final potential energy

$mg(h + x) = 1/2Kx^2$

$2mgh + 2mgx = Kx^2$

$Kx^2 - 2mgx - 2mgh = 0$

$\Rightarrow 1960x^2 - 2 \times 2 \times 9.8 \times x - 2 \times 2 \times 9.8 \times 0.4 = 0$

$\Rightarrow x = 0.1 \text{ m} = 10 \text{ cm}$

24. $U = \left(\frac{x^3}{3} - 4x + 6\right)$.

For equilibrium $F = -\frac{dU}{dx} = 0 \Rightarrow -\frac{3x^2}{3} + 4 = 0 \Rightarrow x^2 = 4\ x = \pm 2$

$\frac{d^2U}{dx^2} = 2x$; $\frac{d^2U}{dx^2} = +$ ve if $x = 2$

$\therefore$ Potential energy is minimum at $x = 2$ m and it is the position of stable equilibrium.

$\frac{d^2U}{dx^2} = -$ ve if $x = -2m$. Therefore potential energy is maximum and it is the position of unstable equilibrium.

25. 1 is correct because at 3 force is opposite to displacement. At 2, u is minimum. 2 is incorrect because at 2 , there is net force hence no equilibrium.

3 is correct because at 1,3 and 2 force is zero. At $b = \frac{du}{dx} = -F = 0$

4 is correct. Positive work is done after position 1, hence K.E will increase. So K.E is least at 1.

26. $f = \frac{dv}{dx}$, Therefore, F = 0 at points B and C.

WORKSHEET – 5
LEVEL – I

1) A	2) C	3) A	4) A	5) C
6) C	7) A	8) A		

LEVEL – II

1) A	2) B	3) D	4) B	5) A
6) A	7) B	8) A	9) D	10) A
11) B	12) C	13) D	14) B	15) C
16) B	17) C	18) B	19) D	20) B
21) B				

LEVEL – III

22) A	23) A	24) B	25) D	
26) a-2; b-3; c-4; d-1		27) 10	28) ABCD	29) 1
30) ABCD	31) 6			

SOLUTIONS

LEVEL – II

1. If the maximum height reached by the body is H. On reaching it its, P.E = mgH This should be equal to the total energy at any instant.
   i.e., $mgH = 400 + 580 = 980;\ 2 \times 10 \times H = 980; \Rightarrow H = 49$ m
2. Given the P.E of the body = K.E of the body $mg10 = \frac{1}{2}mv^2$
   or $v^2 = 2g \times 10 = 2 \times 9.8 \times 10 = 196$ or $v = 14\ ms^{-1}$
3. P.E lost by $slab = P.E$ stored in the spring, $Mg(H + y) = \frac{1}{2}Ky^2$.
4. Law of conservation of energy
5. $E = mc^2;\ E = \frac{100}{1000} \times (3 \times 10^8)^2 = 9 \times 10^{15}$ J
6. $\frac{PE}{KE} = \frac{mgH}{\frac{1}{2}mv^2}; \frac{5}{2} = \frac{9.8 \times 400}{v^2}$
   $v = 56$ m/s
7. ASC to law of conservation of energy
8. According to the principle of conservation of energy,
   $\frac{1}{2}Mv^2 = Mgh$ or $v = \sqrt{2gh}$ here, $h = 10$ cm, $g = 980 cms^{-2}$
   $\therefore\ v = \sqrt{2 \times 980 \times 10} = 140\ cm\ s^{-1}$
9. $h = [1 - \cos\theta]$, loss in P.E = Gain in K.E, K.E = mgh = $mgl(1 - \cos\theta)$
   $$= \frac{100}{1000} \times 10 \times 1\left(1 - \frac{1}{2}\right) = 0.5\ J$$
10. By law of conservation of energy $= \frac{1}{2}Kx^2 = \frac{1}{2}mv^2, = \frac{1}{2} \times 4 \times (2)^2, \frac{1}{2} \times 100 \times x^2$
    $100x^2 = 16, x = 0.4m$
11. $KE = mgl(1 - \cos\theta)$
12. loss in PE = gain in KE
    $mgR = \frac{1}{2}mv^2; v^2 = 2gR; v = \sqrt{2gR}$
13. $t_d = \sqrt{\frac{2h}{g}};\ 4 = \sqrt{\frac{2h}{g}}; 16 = \frac{2 \times h}{10}$
    $h = 80$ m
    After 1sec body is at a height of 75 m from ground
    $\therefore$ % of potential energy still remained is

$= \frac{\text{mg(h)}}{\text{mg(H)}} \times 100 = \frac{75}{80} \times 100 = 93.75\%$

14. $\text{mgL} = \frac{1}{2}\text{MV}^2$

15. $v^2 - u^2 = 2as; v^2 - v_0^2 = 2 \times g \times 10;$
$v\sqrt{v_0^2 + 20g}$, K.E at $B = \frac{1}{2}m(v_0^2 + 20g)$
since 50% energy is lost energy left is $= \frac{1}{2}\ \text{m}\left(\frac{\text{v}_0^2 + 20\ \text{g}}{2}\right) = \text{mg} \times 10$
$\Rightarrow \text{v}_0^2 + 20\ \text{g} = 40\ \text{g}, \text{v}_0 = \sqrt{20\ \text{g}} = 14\ \text{ms}^{-1}$

16. At any point ME is constant
$\therefore ME = 580 + 1380$
ME = 1960
ME at point of projection also 1960
$\therefore$ At point of projection ME = KE
$$1960 = \frac{1}{2}\text{mv}^2;\ 1960 = \frac{1}{2} \times 10\text{v}^2;\ 196 \times 2 = \text{v}^2;\ \text{v} \approx 19.8\ \text{m/s}$$

17. $\frac{1}{2}mu^2 = KE + PE$

18. when block B descends by a distance ' l ', pulley p ascends by a distance $\frac{1}{2}$ and block A ascends by a distance $\frac{1}{2}$. If v is the speed of block A (up), $2v$ would
be the speed of block B (down) Net loss in P.E of system of block B- Gain in P.E of block A. $mgl - mg\frac{1}{2} = \frac{mgl}{2}$. Total gain in $K.E = \frac{1}{2}mv^2 + \frac{1}{2}m(2v)^2 = \frac{5}{2}mv^2$, loss in P.E = gain in K.E $\text{mg}\frac{1}{2} = \frac{5}{2}mv^2 \Rightarrow v^2 = \frac{gl}{5}, v = \sqrt{\frac{gl}{5}}$

19. Total energy of projectile at top of tower $= mgh + \frac{1}{2}mu^2$
Total energy of projectile on reaching the ground $\frac{1}{2}\text{mv}^2$.
From conservation of energy $\frac{1}{2}mv^2 = mgh + \frac{1}{2}mu^2 \cdot v = \sqrt{u^2 + 2gh}$.

20. $x = \frac{2g}{K}$; from conservation of mechanical energy $mgx = \frac{1}{2}Kx^2 + \frac{1}{2}mv^2$

21. The initial extension in spring $x_0 = \frac{mg}{K}$.
J ust after collision of B with A, the speed of the combined mass is $\frac{V}{2}$. For the spring to just attain natural length the combined mass must rise up $x_0 = \frac{mg}{K}$ and come to rest. Applying conservation of energy.

23. Decrease in gravitational P.E of $B = m_B gh, 2 \times 10 \times 1 = 20$ J .

24. Increase in gravitational P.E of $A = m_A gh, 1 \times 10 \times 1 = 10$ J.

25. $m_B gh = m_A gh + \frac{1}{2}m_A v^2 + \frac{1}{2}m_B v^2, 2 \times 10 \times 1 = 1 \times 10 \times 1 + \frac{1}{2} \times 1 \times v^2 + \frac{1}{2} \times 2 \times v^2$
$20 = 10 + 0.5\text{v}^2 + \text{v}^2, 1.5\text{v}^2 = 10, \text{v} = 2.58\ \text{ms}^{-1}$.
$\text{KE}_\text{B} = \frac{1}{2}mv^2 = \frac{1}{2} \times 2 \times (2.58)^2 = 6.67$ J

27. $\text{mgH} = 3.6 + 6.2 = 9.8$ J, $0.1 \times 9.8 \times \text{H} = 9.8$ J, H = 10 m.

28. Here, initial height of the ball, $h_1 = 100$ m
initial velocity of the ball, $v_1 = 0$ final height of the ball, $h_2 = 20$ m
Let v_2 be the velocity acquired by the ball on attaining height h_2. Then according to the law of conservation of energy. $Mgh_1 + \frac{1}{2}Mv_1^2 = Mgh_2 + \frac{1}{2}Mv_2^2$
Since $v_1 = 0$, we have $\text{V}_2 = \sqrt{2g(h_1 - h_2)} = \sqrt{2 \times 10 \times (100 - 20)} = 40\ \text{ms}^{-1}$

29. $mg(h+x)=\frac{1}{2}kx^2$ substitute values and solving $x=0.1m$.

30. Loss of GPE = mgh = $1\times10\times1=10$ J, K.E of 1 kg block = $\frac{1}{2}mv^2=\frac{1}{2}\times1\times(0.3)^2=0.045$ J work done against friction = $\mu mgs=\mu\times4\times10\times2=80\mu$J Loss on GPE-work done against friction = K.E of two blocks $10-80\mu=0.045+\frac{1}{2}\times4\times(0.6)^2; 10-0.045-80\mu=\frac{1}{2}\times4\times(0.6)^2$, solving $\mu=0.12$

31. $(K.E+P.E)_A=(K.E+P.E)_B=6$ m/s

WORKSHEET – 6

1) B 2) C 3) B 4) d

LEVEL – II

1) B 2) D 3) C 4) B 5) A

6) A 7) B 8) D 9) B 10) D

11) C 12) A 13) C 14) B 15) B

16) D 17) A 18) C 19) C

LEVEL – III

20) A, B, C 21) A 22) C 23) D 24) D

25) a-1; b-5; c-2,3; d-4 26) D 27) B 28) C

29) a-2; b-3; c-4; d-1 30) 4

SOLUTIONS

LEVEL – II

2. $p=\frac{w}{t}=\frac{mgh}{t}\Rightarrow t=\frac{mgh}{p}=\frac{200\times10\times40}{10\times1000}=8$sec

3. $W=Pt$

4. $P=\frac{(m+M)gh}{t}$

5. $P=\vec{F}\cdot\vec{V}$

6. $P=\frac{1}{2}\frac{nm}{t}v^2$

7. $P=\vec{F}\cdot\vec{V}$

8. $p=\frac{1}{2}\frac{mnv^2}{t}$

9. $P=\frac{Mgh+\frac{1}{2}Mv^2}{t}$

10. $P=n\times\frac{1}{2}mv^2/t=\frac{nmv^2}{2t}=5.4$ kW

11. $P=\frac{nmu^2}{t}, P\alpha mu^2$

12. $P=3t^2-2t+1;$

$$w=\int pdt=\int_2^4(3t^2-2t+1)=\left[\frac{3t^3}{3}-\frac{2t^2}{2}+t\right]_2^4$$

$$[((4)^3-(4)^2+4)-(2)^3-(2^2)+2]=[64-16+4-8+4-2]=46]$$

19. $P=Fv=M\frac{dv}{dt}v$ Hence, $vdv=\frac{P}{m}dt$ On integration, we find $v\propto\sqrt{t}$

LEVEL – III

20. $w = \int pdt = \int (2t+2)dt = \left[2 \times \frac{t^2}{2} + 2t\right] = t^2 + 2t$
$= (2)^2 + 2(2) = 4 + 4 = 8$ J

24. $p = \frac{dw}{dt} = \frac{d}{dt}(2t^2) = 4t;\ p = 4t;\ 20 = 4t;\ t = 5$ s

25. $P = F \cdot v = (2\hat{\imath} + 2\hat{\jmath}) \cdot (2\hat{\imath}) = 4$ watt

26. $P = F \cdot V = (4\hat{\imath} + \hat{\jmath})(2x\hat{\imath} + \hat{\jmath}) \Rightarrow 9 = 8x + 1 \Rightarrow 8x = 8 \Rightarrow x = 1$

27. $P = F \cdot v = (ai + 2j) \cdot (\hat{\imath} + \hat{\jmath}) \Rightarrow a + 2 = 10 \Rightarrow a = 8$

28. $p = \frac{mgh}{t} \Rightarrow t = \frac{mgh}{p} = \frac{400 \times 10 \times 20}{20 \times 10^3} = 4$sec

WORKSHEET – 7

1) A	2) B	3) D	4) B	5) A
6) A	7) D	8) C		

LEVEL – II

1) B	2) D	3) B	4) B	5) D
6) B	7) C	8) C	9) B	10) B
11) C	12) B	13) D	14) C	15) C
16) C	17) D	18) C	19) A	20) B

LEVEL – III

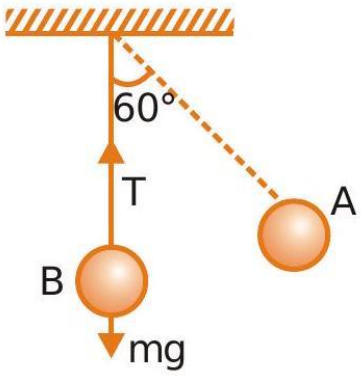
21) A, D	22) C	23) A	24) C	25) D
26) a-r; b-s; c-p,q; d-p,q		27) 80	28) 180	29) 60^0
30] a-3;b-2;c-4;d-1				

SOLUTIONS

LEVEL – II

1. Magnitude of velocity is changing continuously.

2. For the bob, from conservation of energy, $mgl(1-\cos\theta) = \frac{1}{2}mv^2$



$v = \sqrt{2gl(1-\cos\theta)} = \sqrt{2gl\left(1-\frac{1}{2}\right)} = \sqrt{gl}$

At B, if T tension in the string.

$T - mg = \frac{mv^2}{l} = \frac{mgl}{l} = mg \Rightarrow T = 2mg = 2 \times 2 \times 9.8 = 39.2\ N$

3. For the maximum tension in the string, the object should be at the lowest position of the vertical circle. Then $T = mg + \frac{mv^2}{r} = 2 \times 10 + \frac{(2\times16)}{1} = 52\ N$

4. As the bob is at the lowest position of vertical circle, critical velocity that should be given to it is $\sqrt{5gl}$.
Where I is string length
$V = \sqrt{5 \times 9.8 \times 1} = 7\ ms^{-1}$

5. $\frac{1}{2}mv^2 = mgl \Rightarrow v = \sqrt{2gl}$

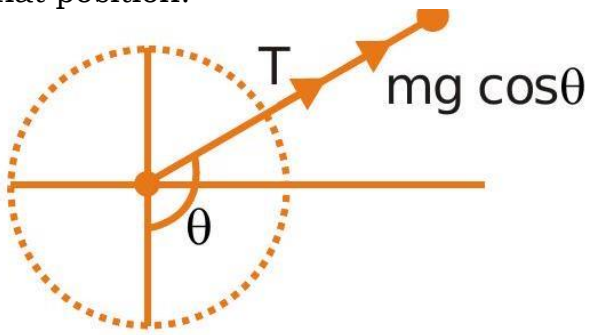
6. At the lowest point on a vertical circle, the minimum required velocity $v = \sqrt{5rg}$
$\Rightarrow v \propto \sqrt{r}$
$\therefore \frac{v_1}{v_2} = \sqrt{\frac{r_1}{r_2}}$ but $r_2 = \frac{r_1}{4}$
$\therefore v_2 = \frac{v_1}{2} = \frac{v}{2}$

7. $\frac{1}{2}m(5gR) = \frac{1}{2}mv^2 + mgR(1 - \cos\theta)$
$5gR = v^2 + 2gR(1 - \cos\theta)$;
$v = \sqrt{gR(3 + 2\cos\theta)}$

8. Let final speed of the bob = v at the given condition from law of conservation of energy

$$\frac{1}{2}m[u^2 - v^2] = mgh[1 - \cos\theta]; 3gl - v^2 = 2gl[1 - \cos\theta]$$
$$v^2 = 3gl - 2gl[1 - \cos\theta]; v^2 = gl[3 - 2[1 - \cos\theta]]$$

Draw F.B.D. at that position.



$T - mg\cos\theta = \frac{mv^2}{r}$ (Since the string becomes slack, T = 0]
$-mg\cos\theta = \frac{m[gl[3 - 2[1 - \cos\theta]]]}{l}$
$-\cos\theta = 3 - 2 + 2\cos\theta, -3\cos\theta = 1$
$\cos\theta = -\frac{1}{3} \Rightarrow \theta = \cos^{-1}\left(-\frac{1}{3}\right)$.

9. $mg - mg\cos\theta = \frac{mv^2}{l}; v^2 = gl(1 - \cos\theta)$
Applying conservation of energy

$$\frac{1}{2}mgl = \frac{1}{2}mgv^2 + mgl(1 + \cos\theta)$$
$$v^2 = gl - 2gl(1 - \cos\theta)$$

Solving (1) and (2) $v^2 = gl(1 - \cos\theta)$
So, $= \sqrt{\frac{gl}{3}} = \sqrt{\frac{9.8\times15}{3}} = 7\ ms^{-1}$.

10. $mgh = \frac{1}{2}m(5rg) \Rightarrow h = \frac{5r}{2} = \frac{2.5}{2} = 1.25m$

11. $T = \frac{mv^2}{r} + mg = 7mg; \frac{mv^2}{r} = 6mg$;

$$r = \frac{v^2}{6g}$$

12. Tension $T = mg + \frac{mv^2}{r} = mg + \frac{m}{l}[2gl(1-\cos\theta)]$

$= mg + 2mg(1-\cos\theta) = mg + \frac{2mg}{2} = 2mg = 2(50\times10^{-3})(10) = 1N$

13. $\frac{V_1}{V_2} = \sqrt{\frac{3+2\cos 60°}{3+2\cos 120°}} = \sqrt{\frac{3+2\left(\frac{1}{2}\right)}{3+2\left(-\frac{1}{2}\right)}} = \frac{2}{\sqrt{2}} = \sqrt{2}:1$

14. $T_1 = \frac{mv^2}{r} - mg; T_2 = \frac{mv^2}{r} + mg.$

Then $T_2 - T_1 = 2mg$

15. At the highest point, $V \geq \sqrt{gR}$

16. $r = \frac{V^2}{(n-1)g}$ where $W_{\max} = nmg$

17. If θ is angular amplitude at mean position $T = mg(3-2\cos\theta)$

18. $u^2 = 5gR$; $v^2 = u^2 - 2gR = 5gR - 2gR = 3gR$

Tangential acceleration at B is $a_t = g(\downarrow)$

Centripetal acceleration at B is $a_c = \frac{v^2}{R} = 3g(\leftarrow)$

Total acceleration at B $a = \sqrt{a_t^2 + a_c^2} = g\sqrt{10}$

20. While crossing the equilibrium position

$$T = mg + \frac{mv^2}{r} = mg + \frac{m}{l}[2gl(1-\cos\theta)]$$

$$= mg + 2mg - 2mg\cos\theta = \mu\gamma[3-2\cos\theta]$$

$$\therefore 3 - 2\cos\theta = \frac{2940}{(150)(9.8)} = 2$$

$$\Rightarrow 2\cos\theta = 1 \Rightarrow \cos\theta = \frac{1}{2} \Rightarrow \theta = 60°$$

LEVEL – III

21. Velocity at bottom is 10 $ms^{-1} = \sqrt{5\times10\times2} = \sqrt{5gR}$

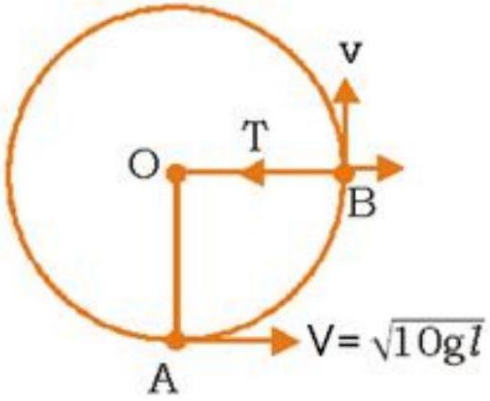
So, the velocity at highest point is $\sqrt{gR} = \sqrt{20} = 2\sqrt{5}$ ms^{-1}

22. Conceptual; tension at highest point is zero.

23. When the string is horizontal tension, = ?

Let v is the final velocity of the bob.

According to law of conservation of energy



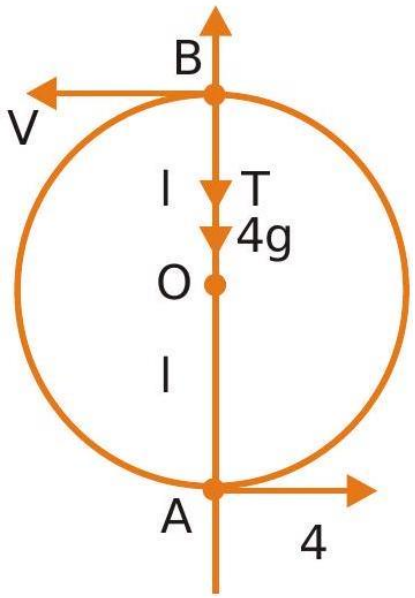
$$\frac{1}{2}m(v^2-u^2)=-mgh\cdot\frac{1}{2}[v^2-10gl]=-gl\ v^2-10gl=-2gl$$
$$v^2=-2gl+10gl=8gl$$

Draw F.B.D of a bob at the position shown $T=\frac{mv^2}{r}=\frac{m\times 8gl}{l}=8mg$

24. When the bob in at the highest point

From law of conservation of energy $\frac{1}{2}m[v^2-u^2]=-mg\times 21$

$v^2-10g\ |=-g\times 4I\ v^2=6gl$

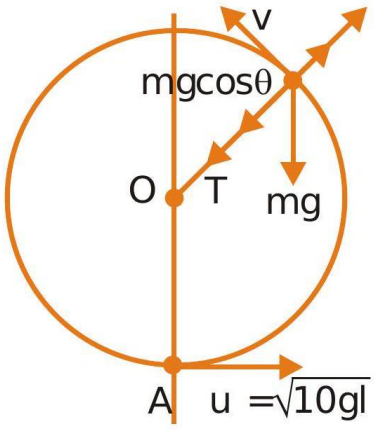


Draw F.B.D. for a bob at the highest point shown $T+mg=\frac{mv^2}{r}$

$\mathrm{T}=\frac{\mathrm{mv}}{\mathrm{r}}-\mathrm{mg}$

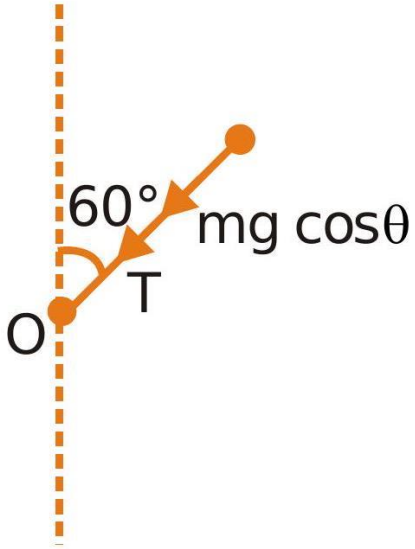
From (1) & (2) $\mathrm{T}=\frac{\mathrm{m6gl}}{\mathrm{l}}-\mathrm{mg}=6\mathrm{mg}-\mathrm{mg}=5\mathrm{mg}$

25. Again from L.C.E,



$$\frac{1}{2}m[v^2-u^2]=-mg[I+I\sin 30°]$$
$$v^2-10gl=-2gl\left[I+\frac{1}{2}\right]$$
$$v^2-10gl=-3gl\ v^2=-3gl+10gl=7gl$$

Draw F.B.D. for a bob at the position B



$T + mg\cos\theta = \frac{mv}{r}$

$T = \frac{mv}{r} - mg\cos 60^\circ = \frac{m \times 7gl}{l} - mg \times \frac{1}{2} = mg\frac{[14-1]}{2} = \frac{13mg}{2} = 6.5mg$

27. $T = \frac{mv^2}{r} - mg = \frac{2\times5\times5}{50\times10^{-2}} - 2 \times 10 = 100 - 20 = 80$ N

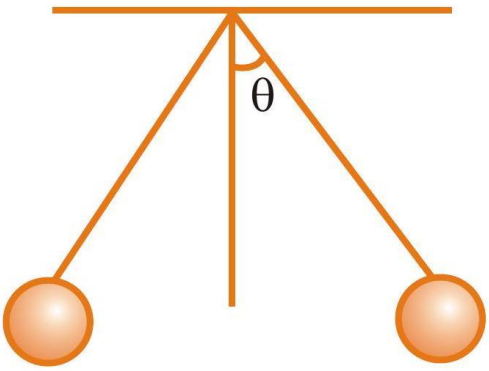
28. Conceptual

29. At lowest point $T = mg + \frac{mv^2}{r}$

$\because mgh = \frac{mv^2}{2}$

$\Rightarrow mgr\cos\theta = \frac{mv^2}{2} (\because h = r\cos\theta)$

$\Rightarrow v^2 = 2gr\cos\theta$



$2mg = mg + \frac{m}{r}(2gr\cos\theta);$

$mg = 2mg\cos\theta;\ \cos\theta = \frac{1}{2} \Rightarrow \theta = 60^\circ$

30. After vehicle is stopped, let the velocity acquired by the bob is v. $h = I - I\cos 60 = 2.5$ m; $mgh = \frac{1}{2}mv^2$; $v = \sqrt{2gh} = 5\sqrt{2}$ ms^{-1} Net force at lowest point, $\frac{v^2}{T} = 10$ ms^{-2}; at point B. $T = mg\cos 60^\circ$

Net force $= mg\sin 60 = 10\sqrt{3}$ N; acceleration $= g\sin 60 = 5\sqrt{3}$ ms^{-2}

2.CENTRE OF MASS & COLLISIONS

WORKSHEET-1

LEVEL – I

1.C 2. C 3. B 4. A 5. C 6. D 7. D

8. B

LEVEL – II

1. D 2. C 3. C 4. B 5. B 6. B 7. D

8. B 9. B 10. C 11. B 12. A 13. B 14. D

15. D 16. C 17. B

LEVEL – III

18. C 19. C 20. D 21. A, B, C, D

22. $a-s; b-r; c-q; d-p$

23. A 24. B 25. B 26. $y=5$ 27. B 28. 1, 22

29. a-2;b-1;c-3;d-4

SOLUTIONS

LEVEL – II

1. Take origin at 5 kg mass.

$$X_{\text{Com}} = \frac{5(0)+4(0.3)+6(1)}{5+4+6}$$

$$= \frac{1.2+6}{15} = \frac{7.2}{15} = 0.48 \text{ m}$$

Ans: (D)

2. Take ' O ' as Origin.

X-Coordinate of com of first half plate $\frac{1}{4}$

X-Coordinate of com of second half plate $\frac{31}{4}$

$$X_{\text{com}} = \frac{\rho\left(\frac{1}{4}\right)+3\rho\left(\frac{31}{4}\right)}{\rho+3\rho} = \frac{\frac{1}{4}+\frac{91}{4}}{4} = \frac{101}{16} = \frac{51}{8}$$

Ans (C)

3. $S = \frac{m \times L}{m+6m}$

$$S = \frac{mL}{7\ m}; S = \frac{L}{7}$$

Ans: (C)

4. shift=radius

$$2\pi r = \pi;\ r = \frac{1}{2};\ r = 0.5m$$

Ans: (B)

5. $$\text{shift} = \frac{m_{\text{removed}}\ (d)}{\text{Total mass} - \text{removed mass}}$$

Where d = distance of COM of removed mass from Original COM

$$= \frac{\rho \times \pi r^2 \times t(R-r)}{\rho \times \pi R^2 \times t - \rho \times \pi r^2 \times t}$$

$$= \frac{r^2(R-r)}{R^2 - r^2} = \frac{r^2}{R+r}$$

$$= \frac{(20)^2}{50+20} = \frac{400}{70} = 5.7$$

Ans: (B)

6. $\vec{r}_{COM} = \frac{m_1\vec{r}_1 - m_2\vec{r}_2}{\vec{r_1} - \vec{r_2}}$

Ans: (B)

7. COM is point of intersection of diagonals

$$\left(\frac{L}{2}, \frac{W}{2}\right)$$

Ans: (D)

8. $m_1x_1 = m_2x_2$

$m_1dx_1 = m_2dx_2$

$10 \times 2 = 20 \times dx_2$

$20 = 20 \times dx_2$

$dx_2 = 1$ cm

1 cm Right

Ans: (B)

9. $X_{cm} = \frac{1(0)+2(2)+3(1)}{1+2+3} = \frac{4+3}{6} = \frac{7}{6}$ cm

$$Y_{cm} = \frac{1(0) + 2(0) + 3(\sqrt{3})}{1+2+3}$$

$$= \frac{3\sqrt{3}}{6} = \frac{\sqrt{3}}{2}$$

Distance $= \sqrt{\left(\frac{7}{6}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}$

$= \sqrt{\frac{49}{36} + \frac{3}{4}} = \sqrt{\frac{19}{9}} = \sqrt{2}$ m (approximately)

Ans: (B)

10. $X_{cm} = 0$

$$\frac{\frac{m}{3}(40) + \frac{m}{3}(20) + \frac{m}{3}(x_3) = 0}{m}$$

$40 + 20 + x_3 = 0;\ x_3 = -60$ cm

$$Y_{cm} = 0; \frac{\frac{m}{3}(0) + \frac{m}{3}(-60) + \frac{m}{3}(y_3)}{m} = 0$$

$0 - 60 + y_3 = 0;\ y_3 = +60$ cm

$(x_3, y_3) = (-60\ \text{cm}, +60\ \text{cm})$

Ans: (C)

11. shift $= \frac{m_{\text{removed}} \times d}{M_{\text{total}} - m_{\text{removel}}}$

$$= \frac{\frac{M}{4} \times \frac{a}{2}}{M - \frac{M}{4}} = \frac{\frac{M}{4} \times \frac{a}{2}}{\frac{3m}{4}} = \frac{a}{6}$$

Co-ordinates of the lamina are $\left(-\frac{a}{6}, 0\right)$

Ans: (B)

12. $X_{cm} = \frac{m(l)+2\ m(2l)+3\ m(3l)+----+nm(nl)}{m+2\ m+3\ m+------+nm}$

$$= \frac{ml[1 + 2^2 + 3^2 + ------ + n^2]}{m[1 + 2 + 3 + ------ + n]}$$

$$= 1\frac{\frac{n(n+1)(2n+1)}{6}}{\frac{n(n+1)}{2}}$$

$$= I\frac{(2n+1)}{6} \times \frac{2}{1} = \frac{1}{3}(2n+1)$$

Ans:(A)

13. co-ordinates of COM $\left(\frac{6}{5}, \frac{7}{10}\right)$

$$X_{cm} = \frac{6}{5}$$

$$\frac{m(0) + 2m(2) + M_1(2) + M_2(0)}{m + 2m + m_1 + m_2} = \frac{6}{5}$$

$$5(4m + 2M_1) = 6(3m + M_1 + M_2)$$

$$20m + 10M_1 = 18m + 6M_1 + 6M_2$$

$$4M_1 - 6M_2 + 2m = 0 ----- (1)$$

$$Y_{cm} = \frac{7}{10}$$

$$\frac{m(0) + 2m(0) + M_1(1) + M_2(1)}{m + 2m + M_1 + M_2} = \frac{7}{10}$$

$$10(M_1 + M_2) = 7(3m + M_1 + M_2)$$

$$3M_1 + 3M_2 - 7m = 0 ----(2)$$

$$M_1 + M_2 = 7m$$

Solving (1) & (2); $M_1 = 4m, M_2 = 3m$

Ans: (B)

14. shift $= \frac{m_{\text{removed}} \times d}{M_{\text{total}} - m_{\text{removed}}}$

$$= \frac{\frac{1}{2} \times 20 \times 30 \times (20)}{(20 \times 60) - \left(\frac{1}{2} \times 20 \times 30\right)}$$

$$= \frac{\frac{1}{2} \times 20}{2 - \frac{1}{2}} = \frac{\frac{20}{2}}{\frac{3}{2}} = \frac{20}{3}\ \text{cm}$$

$= +6.67$ cm to the left

i.e. shift $= -6.67$ cm

Ans: (D)

15. $r_1 = \frac{m_2\ d}{m_1 + m_2}$; $r_1 = \frac{16 \times 1.2 \times 10^{-10}}{1.2 + 16}$

$$r_1 = 0.68 \times 10^{-10}\ \text{m}$$

Ans: (D)

16. $Y_{cm} = \frac{m_1 y_1 + m_2 y_2}{m_1 + m_2}$

$$Y_{\text{cm}} = \frac{m\left(\frac{1}{2}\right) + m(L)}{m + m} = \frac{3L}{4}$$

Ans: (C)

17. Distance of COM

From point ' 0 ',

$$\cos\frac{\theta}{2} = \frac{x}{\left(\frac{L}{4}\right)};\ x = \frac{L}{4}\cos\frac{\theta}{2}$$

LEVEL – III

Ans: (B)

18, 19, 20.

$$\text{shift} = \frac{m_{\text{removed}} \times \phi}{M_{\text{total}} - m_{\text{removed}}} = \frac{\left(\frac{ab}{4}\right) \times \left(\frac{\sqrt{a^2 + b^2}}{4}\right)}{ab - \frac{ab}{4}} = \frac{\sqrt{a^2 + b^2}}{12}$$

$$= \sqrt{\left(\frac{a}{12}\right)^2 + \left(\frac{b}{12}\right)^2}$$

$$X_{cm} = \frac{-a}{12}, Y_{cm} = \frac{-b}{12}, Z_{cm} = 0$$

21. A, B, C, D

22. $a \to s, b \to r, c \to q, d \to p$

23. $x_{cm} = \frac{m(0) + 2m(a) + 3m(a) + 4m(0)}{m + 2m + 3m + 4m}$

$$x_{cm} = \frac{2a + 3a}{10};\ x_{cm} = \frac{a}{2}$$

$$Y_{cm} = \frac{m(0) + 2m(0) + 3m(a) + 4m(a)}{m + 2m + 3m + 4m} = \frac{7a}{10}$$

$$(X_{\text{cm}}, Y_{\text{cm}}) = \left(\frac{\text{a}}{2}, \frac{7\text{a}}{10}\right)$$

24. $\text{X}_{\text{cm}} = \frac{(0.5)(0) + 1(4) + 1.5(0)}{3}$

$$= \frac{4}{3} = 1.3 \text{ cm}$$

$$Y_{\text{cm}} = \frac{0.5(0) + 1(0) + 1.5(3)}{3}$$

$$= 1.5 \text{ cm}$$

Ans: (B)

25. $\frac{1(x_1) + 2(x_2) + 3(x_3)}{6} = 2$

$$\Rightarrow \text{x}_1 + 2\text{x}_2 + 3\text{x}_3 = 12$$

Similarly $\text{y}_1 + 2\text{y}_2 + 3\text{y}_3 = 12$

$z_1 + 2z_2 + 3z_3 = 12$

$x_1 + 2x_2 + 3x_3 + 4x_4 = 0$

$4x_4 = -12$

$x_4 = -3$

$y_4 = -3$

$z_4 = -3$

$(x_4, y_4, z_4) = (-3, -3, -3)$

Ans: (C)

26. $r_1 = \frac{m_2 d}{m_1 + m_2}$

$r_1 = \frac{120 \times 3 \times 10^{-10}}{160}$

$r_1 = 2.25 \times 10^{-10}$ m

$r_1 = 225 \times 10^{-12}$ m

$225 = 45y$

$y = 5$

27. $\overrightarrow{r_{cm}} = \frac{m_1\overrightarrow{r_1} + m_2\overrightarrow{r_2} + m_3\overrightarrow{r_3}}{m_1 + m_2 + m_3}$

$= \frac{1(5\hat{\imath} + 3\hat{\jmath}) + 2(-4\hat{\imath} + 3\hat{\jmath}) + 3(-2\hat{\imath} + 6\hat{\jmath})}{1 + 2 + 3}$

$= \frac{5\hat{\imath} + 3\hat{\jmath} - 8\hat{\imath} + 6\hat{\jmath} - 6\hat{\jmath} + 18\hat{\jmath}}{6}$

$= \frac{-9\hat{\imath} + 27\hat{\jmath}}{6} = \frac{-3\hat{\imath} + 9\hat{\jmath}}{2} = -1.5\hat{\imath} + 4.5\hat{\jmath}$

Ans: (B)

28. $x_{cm} = \frac{1(0) + 2(1) + 3\left(\frac{1}{2}\right)}{1 + 2 + 3}$

$= \frac{2 + \frac{3}{2}}{6} = \frac{7}{12}$ cm

$Y_{cm} = \frac{1(0) + 2(0) + 3\left(\frac{\sqrt{3}}{2}\right)}{1 + 2 + 3} = \frac{3\sqrt{3}}{2 \times 6} = \frac{\sqrt{3}}{4}$

Ans: (A, B)

29. $a \to 2, b \to 1, c \to 3, d - 4$

$X_{com} = \frac{mL + m\left(\frac{L}{2}\right) + m\left(\frac{5L}{4}\right)}{3m} = \frac{11L}{12}$

WORKSHEET-2

LEVEL – I

1. A	2. B	3. A	4. B	5. C	6. D
7. B	8. A	9. C			

LEVEL – II

1. C	2. A	3. B	4. D	5. B	6. C
7. D	8. B	9. B	10. C	11. B	12. A
13. C					

LEVEL – III

14. A, B	15. A	16. $a-4; b-3; c-2; d-1$	17. A
18. A	19. A	20. A, B, C, D	21.B, C
22. 1 m/s^2	23. A	24. B 25. B	

SOLUTIONS

LEVEL – II

1. $\vec{P}_{cm} = \vec{P}_1 + \vec{P}_2$
$\vec{P}_{cm} = 2(3) + 3(-1)$
$\vec{P}cm = 3kgm/s$
Ans: (C)

2. $\vec{V}_{cm} = \frac{m_1\vec{v}_1 + m_2\vec{v_2}}{m_1+m_2}$
$= \frac{6(5\hat{\imath}) + 4(10\hat{\jmath})}{6+4} = \frac{30\hat{\imath} + 40\hat{\jmath}}{10}$
$\vec{V}_{cm} = 3\hat{\imath} + 4\hat{\mathbf{j}}$
$|\vec{V}_{cm}| = \sqrt{3^2 + 4^2} = 5$ m/s
Ans: (A)

3. $\vec{a}_{cm} = \frac{m_1\vec{a_1} + m_2\vec{a_2}}{m_1+m_2} = \frac{m(0)+m(\vec{a})}{m+m}$
$\vec{a}_{cm} = \frac{\vec{a}}{2}$
Ans: (B)

4. $\vec{V}_{cm} = \frac{m_1v_1 + m_2v_2}{m_1+m_2}$
$= \frac{4(6) + 2(0)}{4+2} = \frac{4(6) + 2(0)}{4+2} = 4$ m/s
Ans: (D)

5. $a_1 = \left(\frac{3m-m}{3m+m}\right)g$
$= \left(\frac{2\ m}{4\ m}\right)g = \frac{g}{2}\ \downarrow$
$a_2 = \frac{-g}{2}\uparrow$
$a_1 = \frac{-g}{2}\hat{\jmath},\ a_2 = +\frac{g}{2}\hat{\jmath}$
$a_{com} = \frac{3m\left(\frac{-g}{2}\right)\hat{\jmath} + m\left(\frac{g}{2}\right)\hat{\jmath}}{3m+m}$
$= \frac{\frac{g}{2}(-3\hat{\jmath} + \hat{\jmath})}{4} = \frac{g}{8}(-2\hat{\jmath}) = -\frac{g}{4}\hat{\jmath}$
Ans: (D)

6. $\vec{V}_{cm} = \frac{m_1\vec{v_1} + m_2\vec{v_2}}{m_1+m_2}$
$= \frac{6(5\hat{\imath} - 2\hat{\jmath} + 10\hat{k}) + 4(10\hat{\imath} - 2\hat{\jmath} + 5\hat{k})}{6+4}$
$= \frac{70\hat{\imath} - 20\hat{\jmath} + 80\hat{k}}{10}$
$= 7\hat{\imath} - 2\hat{\jmath} + 8\hat{k}$
Ans: (C)

7. Since there is no external force COM will not move. Hence Ans: (D)

8. $\overrightarrow{a_2} = -\left(\frac{m_2 - m_1}{m_1 + m_2}\right) g\hat{j}$

$= -10\left(\frac{2-1}{2+1}\right)\hat{j} = \frac{-10}{3}\hat{j}$

$\overrightarrow{a_1} = \frac{10}{3}\hat{j}$

$a_{cm} = \frac{2\left(\frac{-10}{3}\right)\hat{j} + 1\left(\frac{10}{3}\right)\hat{j}}{2+1}$

$= \frac{-10}{9}\hat{j};\ |\vec{a}_{cm}| = \frac{10}{9}$

$s = \frac{1}{2}att^2 = \frac{1}{2} \times \frac{10}{9} \times 4 = 2.22$ cm

Ans: (B)

9. Distance moved by the Block $(l - x)$ [rightside]

Distance moved by the wedge ' x ' [left]

$\Delta x_{cm} = 0$

$\Delta(x_m) = \Delta(x_{10m})$

$m(l - x) = 10m(x)$

$ml - mx = 10mx$

$ml = 10mx + mx$

$ml = x(11m)$

$x = \frac{ml}{11m} = \frac{l}{11} = \frac{2.2}{11}$

$x = 0.2$ cm

Ans: (B)

10. zero

Ans: (C)

11. $v_1 = -1.5\cos 37\hat{\imath} - 1.5\sin 37\hat{\jmath}$

$v_1 = -1.5\left(\frac{4}{5}\right)\hat{\imath} - 1.5\left(\frac{3}{5}\right)\hat{\jmath}$

$v_1 = -1.2\hat{\imath} - 0.9\hat{\jmath}$

$v_2 = -1.5\cos 37\hat{\imath} + 1.5\sin 37\hat{\jmath}$

$v_2 = -\frac{4}{5}\hat{\imath} + \frac{3}{5}\hat{\jmath}$

$v_2 = 0.8i + 0.6j$

$v_4 = 2\cos 37\hat{\imath} - 2\sin 37\hat{\jmath}$

$v_4 = 2\left(\frac{4}{5}\right)\hat{\imath} + 2\left(\frac{3}{5}\right)\hat{\jmath}$

$v_4 = 1.6\hat{\imath} - 1.2\hat{\jmath}$

$v_{cm} = \frac{m_1\overrightarrow{n_1} + m_2\overrightarrow{n_2} + m_3\overrightarrow{n_3} + m_4\overrightarrow{n_4}}{m_1 + m_2 + m_3 + m_4}$

$= \frac{1(-1.2\hat{\imath} - 0.9\hat{\jmath}) + 1.5(-0.8\hat{\imath} + 0.6\hat{\jmath}) + 0.5(3\hat{\imath}) + 1(1.6\hat{\imath} - 1.2\hat{\jmath})}{1 + 1.5 + 0.5 + 1}$

$= \frac{0.7\hat{\imath} - 1.2\hat{\jmath}}{4}$

$= \sqrt{\left(\frac{0.7}{4}\right)^2 + \left(\frac{1.2}{4}\right)^2} == \sqrt{(0.175)^2 + (0.3)^2} = 0.34$ m/s

Ans: (B)

12. $F = 2m \times a_{cm};\ a_{com} = \frac{F}{2m}$

$s = ut + \frac{1}{2}at^2;\ s = \frac{1}{2} \cdot \frac{F}{2m} t^2;\ s = \frac{Ft^2}{4m}$

Ans: (A)

13. Initial velocity of centre of mass

$\overrightarrow{u_{cm}} = \frac{Mu + m10)}{M+m} = \frac{Mu}{M+m}$

At maximum height velocity of centre of mass becomes zero

$v^2 - u^2 = 2as;\ 0^2 - \left(\frac{Mu}{M+m}\right)^2 = -2gh_{\max}$

$h_{\max} = \frac{1}{2g}\left(\frac{Mu}{M+m}\right)^2$

Height of COM from ground $= H + \frac{1}{2g}\left(\frac{Mu}{M+m}\right)^2$

Ans : (C)

LEVEL – III

14. A, B (centre of mass must not accelerate)

15. 1 (both statement A&B are true)

16. $a \to 4, b \to 3, c \to 2, d \to 1$

17. $\overrightarrow{V_{cm}} = \frac{m_1\overrightarrow{v_1} + m_2\overrightarrow{v_2}}{m_1+m_2}$

$= \frac{100(10\hat{\imath} - 7\hat{\jmath} - 3\hat{k}) + 300(7\hat{\imath} - 9\hat{\jmath} + 6\hat{k})}{100+300}$

$= \frac{1}{4}[10\hat{\imath} - 7\hat{\jmath} - 3\hat{k} + 21\hat{\imath} - 27\hat{\jmath} + 18\hat{k}]$

$= \frac{1}{4}[31\hat{\imath} - 34\hat{\jmath} + 15\hat{k}]$

Ans: (A)

18. $\vec{a}_1 = \frac{\overrightarrow{F_1}}{m} = \frac{\hat{\imath}+2\hat{\jmath}}{1} = \hat{\imath} + 2\hat{\jmath}$

$\overrightarrow{a_2} = \frac{\overrightarrow{F_2}}{m} = \frac{2\hat{\jmath} + 3\hat{k}}{2} = \hat{\jmath} + \frac{3}{2}\hat{k}$

$\overrightarrow{a_3} = \frac{\overrightarrow{F_3}}{m} = \frac{3\hat{\imath} + \hat{k}}{3} = \hat{\imath} + \frac{1}{3}\hat{k}$

$\overrightarrow{a_{cm}} = \frac{m_1\overrightarrow{a_1} + m_2\overrightarrow{a_2} + m_3\overrightarrow{a_3}}{m_1+m_2+m_3}$

$= \frac{4\hat{\imath} + 4\hat{\jmath} + 4\hat{k}}{6} = \frac{4}{6}(\hat{\imath} + \hat{\jmath} + \hat{k}) = \frac{2}{3}(\hat{\imath} + \hat{\jmath} + \hat{k})$

Ans: (A)

19. $\frac{2}{3}(\hat{\imath} + \hat{\jmath} + \hat{k})$

$|\overrightarrow{a_{cm}}| = \frac{2}{3}\sqrt{3} = \frac{2}{\sqrt{3}} ms^{-2}$

Ans: (A)

20. $a_{\text{com}} = -\left(\frac{M-m}{M+m}\right)^2 g\hat{\jmath}$

Ans: (A, B, C, D)

21. $\vec{V}_{\text{cm}} = \frac{m_1\overrightarrow{v_1} + m_2\overrightarrow{v_2}}{m_1+m_2} = \frac{1(4\hat{\imath})+2(0)}{1+2} = \frac{4\hat{\imath}}{3}$

Ans: (B) & (C)

22. $F = M_{\text{total}} \times a_{cm}$

$30 = (10 + 20) \times a_{cm}$

$a_{cm} = \frac{30}{30} = 1\ \text{m/s}^2$

23. $a_1 = g\sin 45°\hat{\imath} = \frac{g}{\sqrt{2}}\hat{\imath}$

$a_2 = g\sin 45°\hat{\jmath} = \frac{g}{\sqrt{2}}\hat{\jmath}$

$\overrightarrow{a_{cm}} = \frac{m_1\overrightarrow{a_1} + m_2\overrightarrow{a_2}}{m_1 + m_2} = \frac{m\left(\frac{g}{\sqrt{2}}\right)\hat{\imath} + m\left(\frac{g}{\sqrt{2}}\right)\hat{\jmath}}{m + m}$

$\overrightarrow{a_{cm}} = \frac{g}{2\sqrt{2}}(\hat{\imath} + \hat{\jmath});\ |\overrightarrow{a_{cm}}| = \frac{g}{2\sqrt{2}}\left(\sqrt{1^2 + 1^2}\right)$

$= \frac{g}{2\sqrt{2}}(\sqrt{2}) = \frac{g}{2}$

Ans : (A)

24. $\frac{g}{\sqrt{2}}$ Ans: (B)

25. $\frac{g}{\sqrt{2}}$ Ans: (B)

WORKSHEET-3

LEVEL – I

1. D	2. A	3. A	4. B	5. D	6. A
7. A	8. C	9. B	10. B	11. B	12. A
13.C	14. A	15. A			

LEVEL – II

1. C	2. D	3. A	4. A	5. D	6. A
7. B	8. D	9. B	10. B	11. B	12. A
13. C	14. A	15. A	16. A	17. C	18. B
19. D					

LEVEL – III

20. A, B

21. $a - 1; b - 1; c - 2; d - 4$

22. B 23. A 24. D

25. 200 m/s

26. $a - 1,3,5; b - 1,3,5; c - 1; d - 1,4$

27. A, B, D 28. A, B, C

SOLUTIONS

LEVEL – II

1. $v = \frac{mu}{M} = \frac{20\times10^{-3}\times150}{2} = 1.5\ \text{m/s}$

Ans: (C)

2. Law of Conservation of Linear momentum.

$40 \times 80 = 32 \times V$

$V = \frac{40 \times 80}{32};\ V = 100\ \text{m/s}$

Ans: (D)

3. $mu - (-mu) = 2mu$

Ans: (A)

4. Impulse $= \Delta p = m(v - (-u))$
$= 500 \times 10^{-3}[10 - (-10)]$
$= 10$ kg m/s

Ans: (A)

5. $J = 0 - mu = mu$
$= -1500 \times \left(90 \times \frac{5}{18}\right) = -37500\text{NS}$

6. Apply Law of Conservation of momentum
$mV + 2m(0) = (m + 2m)V^1$
$mV = 3mV^1$

$V^1 = V/3$ Ans: (A)

7. $p = \sqrt{2m \cdot KE}; p \propto \sqrt{KE}$

$$\frac{P_2}{P_1} = \sqrt{\frac{KE_2}{KE_1}}$$

$KE_{1=}100, KE_2 = 400$

$$\frac{P_2}{P_1} = \sqrt{\frac{400}{100}} = 2$$

$\frac{P_2}{P_1} - 1 = 1;$

$$\frac{P_2 - P_1}{P_1} \times 100 = 100$$

Ans: (B)

8. $J = {}_{\Delta}P$
$= m(\vec{v} - \vec{u}) = 2(3\hat{\imath} + 2\hat{\jmath} - 2\hat{\imath} - 3\hat{\jmath})$
$= 2(\hat{\imath} - \hat{\jmath}) = 2\sqrt{2}NS$

Ans: (D)

9. Apply L.C.L.M
$m(v_0\hat{\imath}) + m(-3v_0\hat{\jmath}) + m(5v_0\hat{k}) = 3mV^1$
$v_0\hat{\imath} - 3v_0\hat{\jmath} + 5v_0\hat{k} = 3V^1$
$V^1 = \frac{v_0}{3}(\hat{\imath} - 3\hat{\jmath} + 5\hat{k})$

Ans: (B)

10. Apply L.C.L.M
$o = m(10\hat{\imath}) + m(10\hat{\jmath}) + m\vec{v}$
$\vec{v} = -10\hat{\imath} - 10\hat{\jmath}$
$|\vec{v}| = \sqrt{10^2 + 10^2} = 10\sqrt{2}$

$135°$ below X - axis

11. $64(2) + 36(-1) = (64 + 36)V$
$128 - 36 = 100$ V
$v = 0.92$ km/hr

Ans: (C)

12. $O = 2(10\sqrt{3}\hat{\imath}) + 2(10\hat{\jmath}) + 2(\vec{v})$
$\vec{V} = -10\sqrt{3}\hat{\imath} - 10\hat{\jmath}$
$|\vec{V}| = \sqrt{\left(10\sqrt{3}\right)^2 + 10^2} = 10\sqrt{3+1} = 20$ m/s

Ans: (B)

13. Apply L.C.L.M

$O = 4V + (A-4)V^1$

$V^1 = \dfrac{-4V}{A-4}$

ANS: (D)

14. $\vec{J} = \Delta P$

$-3m\hat{\imath} = m\vec{v} - m(2\hat{\imath} + 3\hat{\jmath})$

$-3\hat{\imath} = \vec{V} - (2\hat{\imath} + 3\hat{\jmath})$

$\vec{V} = -3\hat{\imath} + 2\hat{\imath} + 3\hat{\jmath}$

$\vec{V} = -\hat{\imath} + 3\hat{\jmath}$ m/s

ANS: (A)

15. Let V^1 is the velocity of platform

$\overline{V_{m_1 p}} = \overline{V_m} - \overline{V_p}$

$-V = \overline{V_m} - V^1$

$\overline{V_m} = V^1 - V$

$\overline{V_{cm}} = 0$

$\dfrac{m\overline{v_m} + Mv^1}{m+M} = 0$

$m(v^1 - v) + Mv^1 = 0$

$mv^1 + Mv^1 = mv$

$v^1 = \dfrac{mv}{m+M}$

Ans: (A)

16. Let V be the common velocity

$V = \sqrt{2gh} = \sqrt{2 \times 10 \times 1.8} = 6$ m/s

$mu = (m+M)V$

$50 \times u = (50 + 450)v$

$50 \times u = 500 \times v$

$50 \times u = 500 \times 6$

$u = 60$ m/s

Ans: (A)

17. m(v + u) away from the wall.

18. Let 4 m is the velocity of object

$\vec{v}_1 = v\hat{\imath}, \vec{v}_2 = v\hat{\jmath},; \overrightarrow{v_3} = 1, \overrightarrow{v_{cm}} = 0$

$\dfrac{2m(\vec{v}) + m(v\hat{\imath}) + m(v\hat{\jmath})}{4m} = 0$

$2\vec{v} + v\hat{\imath} + v\hat{\jmath} = 0$

$\vec{v} = -\dfrac{v\hat{\imath} - v\hat{\jmath}}{2}; \ \vec{v} = -\dfrac{v}{2}(\hat{\imath} + \hat{\jmath})$

$|\vec{v}| = \dfrac{v}{2}\sqrt{2}$

$= \dfrac{v}{\sqrt{2}}$ m/s

19. $F = 4 \times 35 \times 10^{-3} \times 400 = 56N$

Ans: (D)

LEVEL – III

20. Only momentum is conserved

Ans: (A, B)

21. $a \to 1, b \to 1, c \to 2, d \to 4.$

22. $\vec{u} = -2\cos 45°\hat{j} + 2\sin 45°\hat{\imath}$

$\vec{u} = \frac{2}{\sqrt{2}}\hat{\imath} - \frac{2}{\sqrt{2}}\hat{j}$

$\vec{u} = \sqrt{2}\hat{\imath} - \sqrt{2}\hat{j}$

$\vec{v} = 2\sin 45^0\hat{\imath} + 2\sin 45^0\hat{j}$

$\vec{v} = \sqrt{2}\hat{\imath} + 2\sqrt{2}\hat{j}$

change in velocity $= \vec{v} - \vec{u} = (\sqrt{2}\hat{\imath} + \sqrt{2}\hat{j}) - (\sqrt{2}\hat{\imath} - \sqrt{2}\hat{j})$

$\vec{v} - \vec{u} = 2\sqrt{2}\hat{j}$

$|\vec{v} - \vec{u}| = 2\sqrt{2}$ m/s

23. $\vec{J} = m(\vec{v} - \vec{u})$

$= 50(\sqrt{2}\hat{\imath} + \sqrt{2}\hat{j} - \sqrt{2}\hat{\imath} + \sqrt{2}\hat{j})$

$= 50(2\sqrt{2}\hat{j}) = 100\sqrt{2}\hat{j}$

$= 100(1.414) = 141.4$ kg m/s.

Ans (A)

24. $F = \frac{\Delta p}{\Delta t} = \frac{141.4}{5\times10^{-3}}$

$= 28.28 \times 10^3$

$= 28.3 \times 10^3$ N

Ans: (D)

25. area under f-t graph gives change in momentum.

$mv - mu = f \times t = \frac{1}{2} \times (16 + 8)20 \times 10^3$

$m(v - u) = 240 \times 10^3$

$1200(v - 0) = 240 \times 10^3$

$v = \frac{240 \times 10^3}{1200}$

$v = 200$ m/s

26. $a \to 1,3,5, b \to 1,3,5, c \to 1, d \to 1,4$

27. apply law of conservation of liner momentum.

$15(10\hat{j}) = 2(20\hat{j}) + 3(5\hat{\imath}) + 10(\vec{v})$

$150\hat{j} = 40\hat{j} + 15\hat{\imath} + 10\vec{v}$

$10\vec{v} = -15\hat{\imath} + 110\hat{j}$

$\vec{v} = -1.5\hat{\imath} + 11\hat{j}$

$|\vec{v}| = \sqrt{(1.5)^2 + 11^2} = \sqrt{121 + 2.25} = 11.1$

Ans (A, C, D)

28. $v = \frac{mu}{M}$

$= \frac{30 \times 10^{-3} \times 330}{10} v = \frac{9900 \times 10^{-3}}{10}$

$v = 0.99$ m/s

Ans (A, B, C)

WORKSHEET- 4

LEVEL – I

1. B 2. A 3. A 4. A

LEVEL – II

1.A 2. B 3. C 4. D

LEVEL – III

5. B 6. C 7. B 8. a-1; b-2; c-2; d-1 9. $x = 6$

SOLUTIONS

LEVEL – II

1. Given → along x-axis

$m_A = m_B = 1$ kg

spring constant $k = 200N/M$

$v_0 = (v_B)_{initial} = 2$ m/s

$F_{net} = 0$

According to conservation of linear momentum $(1 \times 2) + 0 = 2 \times 1 \times V$

$2 = 2v$; $v = 1$ m/s

According to the work-energy theorem

$$\frac{1}{2}(2)v^2 - \frac{1}{2}(1)v_0^2 = \frac{1}{2}k\left(x_i^2 - x_f^2\right)$$

$$v^2 - \frac{v_0^2}{2} = \frac{1}{2} \times 200\left(-x_f^2\right)$$

$$1 - 2 = 100\left(-x_f^2\right); x_f = \frac{1}{10}m$$

$x_f = 10$ cm

Ans: (A)

2. At maximum extension the blocks move with common velocity Apply L.C.L.M

$$m_1u_1 + m_2u_2 = (m_1 + m_2)V$$

$$6(2) - 3(1) = (6 + 3)V$$

$$12 - 3 = 9V$$

$V = 1$ m/s

Apply law of conservation of energy

$$\frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 = \frac{1}{2}(m_1 + m_2)V^2 + \frac{1}{2}kx^2$$

$$\frac{1}{2} \cdot 6 \cdot 4 + \frac{1}{2} \cdot 3 \cdot 1 = \frac{1}{2}(6 + 3)(1)^2 + \frac{1}{2}(200)x^2$$

$$\frac{24}{2} + \frac{3}{2} = \frac{9}{2} + \frac{200}{2}x^2$$

$$27 = 9 + 200x^2$$

$$x = \sqrt{\frac{9}{100}}$$

$x = 0.3m$

$x = 30$ cm

3. At maximum compression the blocks move with common velocity

Apply L.C.L.M

$$m_1u_1 + m_2u_2 = (m_1 + m_2)V$$

$$2(1) + 2(0) = (2 + 2)V$$

$$V = \frac{1}{2}m/s$$

Apply law of conservation of energy

$$\frac{1}{2}m_1v_1^2+\frac{1}{2}m_2v_2^2=\frac{1}{2}(m_1+m_2)V^2+\frac{1}{2}kx^2$$

$$\frac{1}{2}(2)1+\frac{1}{2}(2)0=\frac{1}{2}(2+2)\left(\frac{1}{2}\right)^2+\frac{1}{2}(100)x^2$$

$x=0.1m$

$x=10$ cm

4. $F_{exemal}=0$

$L_i=L_f$

$0=-mv_1+Mv_2$

Here $v_1=v_m$, $v_2=v_M$

According to work-energy theorem

$$\frac{1}{2}kx^2=\frac{1}{2}mv_1^2+\frac{1}{2}Mv_2^2$$

$$kx^2=m\left[\frac{M}{m}v_2\right]^2+Mv_2^2$$

$$v_1=\left(\frac{kM}{m(M+m)}\right)^{\frac{1}{2}}x$$

LEVEL – III

Ans: (D)

5. Conservation of momentum

$P_i=P_f$

($\because v_1=v_2=v$)

$m_1u_1+m_2u_2=(m_1v_1+m_2v_2)$

$m_1v_1=m_2v_2=(m_1+m_2)v$

$1(2)+1(0)=(1+1)v$

$2=2$ V

$v_c=1$ m/s

Ans: (B)

6. $k_i=$ initial kinetic energy $=\frac{1}{2}\ m_1u_1^2$

$$k_i=\frac{1}{2}(1)(2)^2$$

$=\frac{1}{2}\times 4=2$ J

$$k_f=\frac{1}{2}(\ m_1+m_2)v_c^2$$

$$k_f=\frac{1}{2}(1+1)(1)^2=\frac{1}{2}(2)=1\text{ J}$$

Loss in K.E $=2\text{ J}-1\text{ J}$

$=1$ J

Ans: (C)

7. loss in KE which is equal to the gain in PE

$1\text{J} = \frac{1}{2}kx^2$

$1 = \frac{1}{2}50x^2;\ \frac{1}{25} = x^2$

$x = \sqrt{\frac{1}{25}} = \frac{1}{5} = 0.2$

$x = 0.2$ m

$\left(\because K = 50\ \text{N/m},\ x = ?\right)$

Ans: (B)

8. a) velocity of centre of mass $V_{com} = \frac{2(2)+2(0)}{2+2} = 1$ m/s $a \longrightarrow 1$

b) Apply L.C.L.M

$m_1u_1 + m_2u_2 = (m_1 + m_2)V$

$2(2) + 2(0) = (2 + 2)V$

$V = 1m/s$

$b \longrightarrow 2$

c) Apply law of conservation of energy

$\frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 = \frac{1}{2}(m_1 + m_2)V^2 + \frac{1}{2}kx^2$

$\frac{1}{2}\cdot 2\cdot 4 + \frac{1}{2}\cdot 2\cdot 0 = \frac{1}{2}(2+2)(1)^2 + \frac{1}{2}(4)x^2$

$4 = 2 + 2x^2$

$x = 1m;$

$c \longrightarrow 2$

d) $\frac{1}{2}kx^2 = \frac{1}{2}\cdot 4\cdot (1)^2$

$= 2J;\ d \longrightarrow 1$

9. At maximum elongation the blocks move with common velocity Apply L.C.L.M

$m_1u_1 + m_2u_2 = (m_1 + m_2)V'$

$m(-V) + m(V) = (m + m)V'$

$V' = 0$ m/s

Apply law of conservation of energy

$\frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 = \frac{1}{2}(m_1 + m_2)V^2 + \frac{1}{2}kx^2$

$\frac{1}{2}(m)V^2 + \frac{1}{2}(m)V^2 = \frac{1}{2}(m + m)(0)^2 + \frac{1}{2}kx^2$

$x = \sqrt{\frac{2m}{k}}V$

$\frac{x}{3} = 2;\ x = 6$

WORKSHEET-5

LEVEL – I

1.C	2. C	3. A	4. C	5. A	6. B
7. A	8. C	9. A	10. C		

LEVEL – II

1. B	2. C	3. A	4. A	5. A	6. A
7.A	8. 1	9. D	10. B	11. C	12. C
13. D	14. D	15. B	16. A	17. A	18. D
19. D					

LEVEL – III

20. A, B, C

21. $1-r, 2-q; 3-p; 4-s$

22. $x=3$ 23. A, B, C 24. A 25. B 26. B

SOLUTIONS

LEVEL – II

1. $m_1\overrightarrow{v_1} + m_2\overrightarrow{v_2} = 0$

$m_1\overrightarrow{v_1} = -m_2\overrightarrow{v_2}$

$\overrightarrow{v_2} = -\frac{m_1}{m_2}\overrightarrow{v_1} = \frac{-0.5}{1} \times 5$

$\overrightarrow{v_2} = -2.5$ m/s

$v_2 = 2.5$ m/s

Ans: (B)

2. $m_1 v_1 + m_2 \times 0 = (M_1 + m_1)v$

$60 \times 10 = (60+40)v;\ v = \frac{600}{100} = 6$ m/s

loss of kinetic energy $= K_f - K_i$

$= \frac{1}{2} \times 60 \times (10)^2 - \frac{1}{2} \times 100 \times (6)^2$

$= \frac{1}{2} \times 60 \times 100 - \frac{1}{2} \times 100 \times 36$

$= \frac{1}{2} \times 100(60-36) = 50 \times 24 = 1200$ J

Loss of K.E $= 1200$ J

Ans: (C)

3. $mv = (M-m)_1 + mV'$

$V' = \frac{mv - (M - mN_1}{m}$

Ans: (A)

4. $u_1 = u, u_2 = 0$

$v_1 = v, v_2 = V$

$m_1 = M, m_2 = m$

$v_2 = \left(\frac{m_2 - m_1}{m_1 + m_2}\right)u_2 + \left(\frac{2m_1u_1}{m_1 + m_2}\right)$

$V' = \left(\frac{2M}{m+M}\right)u$

Ans: (A)

5. $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$

$1 \times 4 + 2 \times 0 = 1 \times 0 + 2v_2$

$v_2 = 2$ m/s Ans (A)

6. Since, $m_1 > m_2$ the lighter body moves with double the speed of heavier body after collision.

Hence, $v_2 = 10$ m/s

Ans: (A)

7. (5 kg) $\longrightarrow$ 3 m/s
(3 kg) $\longleftarrow$ 2 m/s
(5 kg) $\longrightarrow$ 0 m/s
(3 kg) $\longrightarrow v$
$5 \times 3 - 3 \times 2 = 0 + \text{mv}$
$9 = 3\text{v}$
$\text{v} = 3$ m/s

Ans: (A)

8. $v_1 \left(\frac{m_1 - m_2}{m_1 + m_2}\right) u_1 + \frac{2m_2 u_2}{m_1 + m_2}$
$\text{m}_1 = \text{m}\ \text{m}_2 = 2$ m
$u_1 = v\ u_2 = -2v$
$V_1 = \frac{-v}{3} - \frac{8v}{3} = \frac{-9v}{3} = -3v$
$v_2 = \left(\frac{m_2 - m_1}{m_1 + m_2}\right) u_2 + \frac{2m_1 u_1}{m_1 + m_2}$
$= \left(\frac{2m - m}{3m}\right)(-2v) + \frac{2(m)(v)}{3m}$
$= \frac{-2v}{3} + \frac{2v}{3} = 0.$

Ans: (A)

9. $v_2 = \left(\frac{m_2 - m_1}{m_1 + m_2}\right) u_2 + \left(\frac{2m_1 u_1}{m_1 + m_2}\right)$
$\left(\frac{4-6}{4+6}\right)(-5) + \frac{2 \times 6 \times 10}{6+4} = 13$ m/s

Ans: (D)

10. $m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$
$6 \times u + m_2 \times 0 = 6 \times \frac{u}{2} + m_2 u$
$\text{m}_2 = 3$ kg

Ans: (B)

11. $v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right) u_1 + \left(\frac{2m_2 u_2}{m_1 + m_2}\right)$
$v_2 = \left(\frac{m_2 - m_1}{m_1 + m_2}\right) u_2 + \left(\frac{2m_1 u_1}{m_1 + m_2}\right)$
$\text{v}_1 = -0.6$ m/s, $\text{v}_2 = 2.4$ m/s

Ans: (C)

12. $v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right) u_1 + \left(\frac{2m_2 u_2}{m_1 + m_2}\right)$
$v_2 = \left(\frac{m_2 - m_1}{m_1 + m_2}\right) u_2 + \left(\frac{2m_1 u_1}{m_1 + m_2}\right)$
$m_2 > m_1$
$\text{v}_1 = -\text{u}_1$
$\text{u}_1 = \text{V}_1,\ \text{u}_2 = \text{V}_2 = 0$
$\text{v}_1 = -\text{V}_1; \text{v}_2 = 0$

Ans: (C)

13. $v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right) u_1 + \left(\frac{2m_2 u_2}{m_1 + m_2}\right)$

$m_2 > m_1$
$v_1 = -u_1 + (2u_2)$
$v_1 = -2 + 2(-1)$
$v_1 = -4$ m/s
Ans: (D)

14. $v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right) u_1 + \left(\frac{2\, m_2 u_2}{m_1 + m_2}\right)$
$m_2 > m_1, u_2 = 0$
$v_1 = -u_1$
$v = -10$ m/s,
Ans: (D)

15. According to Law of conservation of linear momentum
Olympiad Class Work Book = IX -Physics (Vol-I)
$mv = 2mv;\ v' = \frac{v}{2}$
$(\text{K.E})\ \text{collision} = \frac{1}{2}(2\ m)\left(\frac{v}{2}\right)^2 = \frac{mv^2}{4}$
$(\text{K.E})\ \text{collision} = \frac{1}{2}mv^2 = K$
$\Delta \text{K.E} = \text{elastic P.E} = K - \frac{K}{2} = \frac{K}{2}$
Ans: (B)

16.
$$v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right) u_1 + \left(\frac{2m_2 u_2}{m_1 + m_2}\right)$$
$$v_1 = \frac{(m_2 - m_1)v}{m_1 + m_2};\ [(K.E)_i] = \frac{1}{2} m_1 v^2$$
$$\left[(K.E)_f\right] = \frac{1}{2} m_1 \frac{(m_2 - m_1)^2 v^2}{(m_1 + m_2)^2}$$
Decrease in K.E of $m_1; \Delta K.E = [(K.E)_i] - \left[(K.E)_f\right]$
$$= \frac{1}{2} m_1 v^2 - \frac{1}{2} m_1 v^2 \left[\frac{(m_2 - m_1)}{m_1 + m_2}\right]^2$$
$$= \frac{1}{2} m_1 v^2 \left[1 - \frac{(m_2 - m_1)^2}{(m_1 + m_2)^2}\right]$$
$$= \frac{1}{2} m v^2 \left[\frac{(m_1 + m_2)^2 - (m_2 - m_1)^2}{(m_1 + m_2)^2}\right]$$
$$= \frac{1}{2} m_1 v^2 \left[\frac{4 m_1 m_2}{(m_1 + m_2)^2}\right]$$

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fractional change in $\text{K.E} = \frac{\Delta \text{K.E}}{\text{K.E}_i}$
Ans: (A)

17. Total number of collisions is 3

18. Since the collission is elastic the total K.E. of the system reains constant. hence, there is no loss in K.E of thr system.

19. After t = 5secV = u + at
$= 100 - (9.8)5 = 51$ m/s
$MV = m_1 v_1 + m_2 v_2;\ (50)(51) = (20)(150) + (30)V$

$v = -15$ m/s

Ans: (D)

LEVEL – III

20. Ans (A, B, C)

21. $1 \rightarrow r, 2 \rightarrow q, 3 \rightarrow p, 4 \rightarrow s$

22. $v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right) u_1 + \left(\frac{2m_2 u_2}{m_1 + m_2}\right)$

$v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right) u_1;\ \frac{u}{5} = \left(\frac{1 - m_2}{1 + m_2}\right) u$

$m_2 = \frac{2}{3} = \frac{2}{x};\ x = 3$

23. $v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right) u_1 + \left(\frac{2\, m_2 u_2}{m_1 + m_2}\right)$

$m_1 = 3$ kg, $m_2 = 2$ kg, $u_1 = 10$ m/s$u_2 = 5$ m/s

$v_1 = 6$ m/s

$v_2 = \left(\frac{m_2 - m_1}{m_1 + m_2}\right) u_2 + \left(\frac{2m_1 u_1}{m_1 + m_2}\right)$

$v_2 = 11$ m/s

since collision is elastic, there is no loss in K.E

24. $M = 3m, V = u\cos 60°\hat{\imath}$

$m_1 = m, v_1 = 200\hat{\jmath};\ m_2 = 2m, v_2 = ?$

$MV = mv_1 + mv_2$

$3m(100u\cos 60^0\hat{\imath}) = m(200\hat{\jmath}) + (2m)\overrightarrow{v_2}$

$150\hat{\imath} = 200\hat{\jmath} + 2\overrightarrow{v_2}$

$\overrightarrow{v_2} = 75\hat{\imath} - 100\hat{\jmath}$

$|\overrightarrow{v_2}| = \sqrt{(75)^2 + (-100)^2};\ \overrightarrow{v_2} = 125m$

Ans: (A)

25. Tan $\alpha = \frac{100}{75} = \frac{4}{3}$;

$\sin\alpha = \frac{4}{5}$

$\alpha = \sin^{-1}\left(\frac{4}{5}\right)$

Ans: (B)

26. $V = u\cos 60°;\ V = 100 \times \frac{1}{2}$

$V = 50$ m/s

Ans: (B)

WORKSHEET-6

LEVEL – I

1.C	2. D	3. D	4. B	5. D	6. D
7.B	8. B				

LEVEL – II

1. A	2. C	3. C	4. C	5. B	6. B
7.D	8. B	9. B	10. B	11. C	12. C
13. C	14. A	15. B	16. B	17. C	18. D

LEVEL – III

19. A, C 20. B 21. A 22. B 23. $1-q; 2-r; 3-s; 4-p$

24. $2m$ 25. a-q;b-p;c-s;d-r a-q;b-p;c-s;d-r 26. D 27. A

28. A 29. B, C

SOLUTIONS

LEVEL – II

1. $V_1 = \left(\frac{m_1 - em_2}{m_1+m_2}\right)u_1 + \frac{(1+e)m_2u_2}{m_1+m_2}$

$$V_1 = \left(\frac{1-\frac{2}{3}\cdot 2}{1+2}\right)(12) + \frac{\left(1+\frac{2}{3}\right)\cdot 2\cdot(-24)}{1+2}$$

$V_1 = -28$ m/s

$$V_2 = \left(\frac{m_2 - em_1}{m_1+m_2}\right)u_2 + \frac{(1+e)m_1u_1}{m_1+m_2}$$

$$V_2 = \left(\frac{2-\frac{2}{3}\cdot 1}{1+2}\right)(-24) + \frac{\left(1+\frac{2}{3}\right)\cdot 1\cdot 12}{1+2}$$

$V_2 = -4$ m/s

Ans: (A)

2. $V_n = e^nV;\ V_4 = e^4V$

$\frac{256}{625}V = e^4V;\ e = 0.8$

3. $e = \sqrt{\frac{h_2}{h_1}};$

$$e = \sqrt{\frac{1.5}{2}} = \sqrt{\frac{15}{20}} = \frac{\sqrt{3}}{2}$$

Ans: (C)

4. L.C.L.M :

$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$

$v + m(0) = 6\times\frac{v}{3} + m\times\frac{v}{3}$

$6v = 2v + \frac{mv}{3};\ (6v-2v) = mv$

$3(4v) = mv;\ m = 12$ kg

Ans: (C)

5. apply L.C.L.M

$\frac{m}{n}V = \left(m+\frac{m}{n}\right)\cdot V';\ V' = \frac{V}{n+1}$

Ans: (B)

6. $e = \sqrt{\frac{h_2}{h_1}};$

$e = \sqrt{\frac{0.64h}{h}};\ e = \sqrt{\frac{64}{100}} = \frac{8}{10} = 0.8;$

$e = 0.8$

Ans: (B)

7. $t = \sqrt{\frac{2H}{g}}\left[\frac{1+e}{1-e}\right]$

$H = 19.6$ m; $g = 9.8$ m/s^2

$e = \frac{1}{2}$; $t = \sqrt{\frac{2 \times 19.6}{9.8}}\left[\frac{1+\frac{1}{2}}{1-\frac{1}{2}}\right]$

$t = \sqrt{4}\left[\frac{\frac{2+1}{2}}{\frac{2-1}{2}}\right] \Rightarrow 2\left[\frac{3}{1}\right]$

$t = 6$sec

Ans: (D)

8. Apply L.C.LM

$mV = (m+M)V^{|}$ (where $V^{|}$is common velocity)

$V' = \left(\frac{m}{m+M}\right)V$

Apply conservation of energy

$(m+M)gh = \frac{1}{2}(m+M)(V')^2$

$(m+M)gh = \frac{1}{2}(m+M)\left(\frac{m}{m+M}\right)^2 V^2$

$V = \frac{(m+M)}{m}\sqrt{2gh}$

Ans: (B)

9. $\frac{h_1}{h_2} = \frac{1}{e^2}$ $h_1 = 144$ cm

$h_2 = 81$ cm

$\frac{144 \text{ cm}}{81 \text{ cm}} = \frac{1}{e^2}$; $e^2 = \frac{81}{144}$

$e = \sqrt{\frac{81}{144}} = \frac{9}{12} = 0.75$

$e = 0.75$

Ans: (B)

10. Let ' m ' be the mass of each ball A and B According to conservation of linear momentum

$mu - mv = mv_2$;

$v_2 = u - v$

Where v_2 is the velocity of ball ' B ' after collision.

(Relative velocity of separation)=e(Relative velocity of approach)

$e = \frac{v_2 - v_1}{u_1 - u_2}$; $v_1 = 0$

$e = \frac{u-v}{u-(-v)}$; $e = \frac{u-v}{u+v}$

From componendo and dividendo method $\frac{u}{v} = \frac{1+e}{1-e}$

Ans: (B)

11. When a freely falling ball strikes the ground with a velocity ' v ' and rebounds with a velocity v_1 then $e = \frac{v_1}{v}$ and after nth rebound $v_n = e^n \cdot v$

Ans (A)

12. Total time taken by the ball to stop bounding

$$t = \sqrt{\frac{2\,h}{g}}\left[\frac{1+e}{1-e}\right]$$

Ans: (C)

13. $e = \sqrt{\frac{h_2}{h_1}}; e = \sqrt{\frac{81\text{ cm}}{100\text{ cm}}} = 0.9$

Ans: (C)

14. $h = e^{2n}H$

Ans: (A)

15. Apply L.C.L.M along X-direction

$$mv\cos\left(\frac{\theta}{2}\right) + mv\cos\left(\frac{\theta}{2}\right) = (2m)\frac{v}{2}$$

$$\cos\left(\frac{\theta}{2}\right) = \frac{1}{2};\ \cos\left(\frac{\theta}{2}\right) = \cos 60^\circ$$

$\theta = 120^\circ$ Ans (b)

16. $0 + 0.2 \times 100 = (9.8 + 0.2)v$

$$v = 2\text{ m/s};\ mgh = \frac{1}{2}mv^2$$

$$h = \frac{v^2}{2g};\ h = 0.2\text{ m}$$

Ans: (B)

17. Apply L.C.L.M

$$20v = (20 + 80)v';\ v' = \frac{v}{5}$$

$$\frac{K \cdot E_i}{K \cdot E_f} = \frac{\frac{1}{2}(20)v^2}{\frac{1}{2}(80+20)\left(\frac{v}{5}\right)^2}$$

$\frac{K \cdot E_i}{K \cdot E_f} = 5:1$ Ans (C)

18. $m\alpha r^3$

$r_1 = r, r_2 = 2r$

$m_1 = m,\ m_2 = 8m$

$8m(81) + m(0) = (8m + m)v$

$v = 72$ cm/s Ans (d)

LEVEL – III

19. $m_1 = m, m_2 = 2m, u_1 = v, u_2 = -2v$

$$v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)u_1 + \left(\frac{2m_2u_2}{m_1 + m_2}\right)$$

$$V_1 = -3v;\ v_2 = \left(\frac{m_2 - m_1}{m_1 + m_2}\right)u_2 + \left(\frac{2m_1u_1}{m_1 + m_2}\right)$$

$V_2 = 0$ Ans (A, C)

20. $v_1 = \left(\frac{m-2m}{m+2m}\right)(\sqrt{2gH}) + \left(\frac{2(2m)(-\sqrt{2gH})}{m+2m}\right)$

$$V_1 = \frac{-5\sqrt{2gH}}{3};\ V_1 | = \frac{5}{3}\sqrt{2gH}$$

Ans: (B)

21. Apply conservation of energy

$$mgH' = \frac{1}{2}mV_1^2;\ H' = \frac{V_1^2}{2g};\ H' = \frac{25H}{9}$$

Ans: (A)

22. $v_2 = \left(\frac{2m-m}{m+2m}\right)(-\sqrt{2gH}) + \left(\frac{2m\sqrt{2gH}}{m+2m}\right)$

Apply law of conservation of energy

$$(2\ \mathrm{m})g\mathrm{H}' = \frac{1}{2}(2\mathrm{mN}_2^2$$

$$H' = \frac{V_2^2}{2g}; H' = \frac{H}{9}$$

Ans: (B)

23. The body reaches a maximum height $h = \frac{u^2}{2g}$, after that the situation is equivalent to the body dropped from the height h

The total distance travelled by the ball $= h + h\left(\frac{1+\mathrm{e}^2}{1-\mathrm{e}^2}\right)$

$$= \frac{u^2}{g(1-e^2)}$$

$1 \longrightarrow q$

The total time by the ball to come to rest $= t_a + \sqrt{\frac{2h}{g}}\left(\frac{1+e}{1-e}\right)$

$$= \frac{\mathrm{u}}{\mathrm{g}} + \frac{\mathrm{u}}{\mathrm{g}}\left(\frac{1+\mathrm{e}}{1-\mathrm{e}}\right) \ = \frac{2\mathrm{u}}{\mathrm{g}(1-\mathrm{e})}$$

$2 \longrightarrow r$

Average speed of the ball $= \frac{\text{total dis tance}}{\text{total time}} = \frac{u^2}{g(1-e^2)} \times \frac{g(1-e)}{2u} = \frac{u}{2(1+e)}$

24. $h_n = e^{2n}h_0;\ h_2 = (0.5)^{2\times2}(32)$

$\mathrm{h}_2 = 2$ m

25. $a \longrightarrow \mathrm{q}, \mathrm{b} \longrightarrow \mathrm{p}, \mathrm{c} \longrightarrow \mathrm{s}, \mathrm{d} \longrightarrow \mathrm{r}$

26. Since the colliision is elastic there is no loss in K.E

Ans (D)

27. After the first collision the bodies will exchange their velocities. after some time the string will become tight . at this time the common velocities of the balls is v. Apply L.C.L.M

$$mu + 2mu = 2mv;\ v = \frac{3u}{2}$$

$$K.E_i = \frac{1}{2}mu^2 + \frac{1}{2}m(2u)^2 = \frac{5mu^2}{2}$$

$$K.E_f = \frac{1}{2}(m+m)\left(\frac{3u}{2}\right)^2 = \frac{9mu^2}{4}$$

Loss in K.E $= K.E_i - K.E_f = \frac{mu^2}{4}$

Ans: (A)

28. For the left ball $\mathrm{J} = \mathrm{m}(\mathrm{v} - \mathrm{u})$

$$J = m\left(\frac{3u}{2} - u\right);\ J = \frac{mu}{2}$$

Ans: (A)

29. velocity of B after collision is

$$V_2 = \left(\frac{m_2 - em_1}{m_1 + m_2}\right)u_2 + \frac{(1+e)m_1u_1}{m_1 + m_2}$$

$$V_2 = \frac{\left(1+\frac{1}{2}\right)mV}{m+m};\ V_2 = \frac{3}{4}V$$

apply L.C.L.M

$$mV + m(0) = mV_1 + m\left(\frac{3V}{4}\right);\ V_1 = \frac{V}{4}$$

Impulse = change in momentum

$$= m\left(\frac{3V}{4}\right) - 0\ = \frac{3mV}{4}$$

Loss in K.E $= \frac{1}{2}mV^2 - \left(\frac{1}{2}m(v/4)^2 + \frac{1}{2}m(3V/4)^2\right) = \frac{3mV^2}{16}$ Ans (B, C)

3.ROTATIONAL MECHANICS

WORKSHEET -1

LEVEL - I

1. C	2. A	3. A	4. D	5. D	6. A	7. C	8. B
9. A	10. D						

LEVEL - II

1. B	2. C	3. D	4. B	5. D	6. C	7. A	8. D
9. B	10. D	11. B	12. A	13. A	14.B	15. A	16. C
17.A	18. C	19. B	20. B				

LEVEL - II

21. A-R, B-Q, C-P, D-S 22. A-R, B-Q, C-S, D-P

23.A, B, C, D	24. C	25. A	26. A	27. D	28. C	29. A
30. C	31. B	32. D				

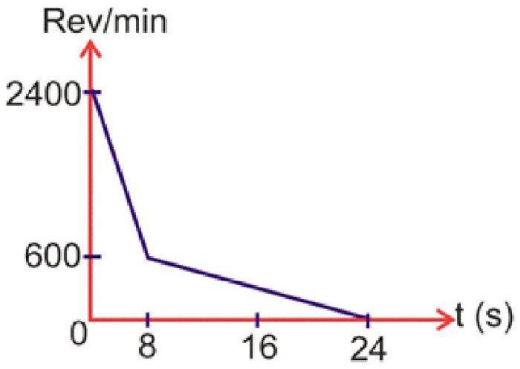
SOLUTIONS

LEVEL - I

1. Angular velocity of all particles of a rigid body in pure rotation is constant. All particles move with some angular velocity
2. $\bar{\omega} \cdot \bar{v} = \omega \vee \cos 90 = 0$
3. In the case of electron revolving around the nucleus, the electro static force between electron and nucleus act as gravitational force
4. The physical quantities which remain constant in uniform circular motion are
   i) Speed (ii) angular velocity (iii) time period (vi) magnitude of centripetal acceleration and (v) magnitude of radius vector so the physical quantities which are variable among the given quantities are : linear velocity and centripetal acceleration
5. By using right hand thumb rule we can say this .
6. Sediments are carried towards the equator so radius of earth is increasing
   $\omega \propto \frac{1}{r}$
7. $\omega = \frac{d\theta}{dt}$, Here $d\theta$ is small and hence it can be represent as vector.
8. Its angular speed that is the rate of change of angular displacement.
9. Angular velocity of seconds hand of watch constant vector.
10. Angular speed same for Both, Because time taken same

LEVEL - II

1.



$$V = 2400\text{rpm} = \frac{2400}{60} = 40\text{rps}$$

Number of Revolutions = area under curve

$$= \frac{1}{2} \times 30 \times 8 + 10 \times 8 + \frac{1}{2} \times 10 \times 16$$

$$= 120 + 80 + 80 \quad N = 280$$

2. $\omega_i = 0$

$\propto = 8\text{rad/s}^2$, $N = 77$

$\theta = \frac{1}{2}\alpha t^2$ $2\pi N = \frac{1}{2}\alpha t^2$

$2 \times \frac{22}{7} \times 77 = \frac{1}{2}(8)t^2$

$(11)^2 = t^2$ $t = 11\text{sec}$

3. $\omega = \frac{\theta}{t} = \frac{\pi}{4}\text{rad/s}$

4. $\omega = \frac{\theta}{t} = \frac{\pi}{10}\text{rad/s}$

5. $v = r\omega$ and $\omega = \frac{2\pi}{T} = 6400 \times 1000 \times \frac{2\pi}{86400} = \frac{64 \times 6.28}{864}$ km/s $= 0.465$ km/s

6. $\omega = \frac{2\pi}{T}\text{rad/s}$

7. $\omega = \frac{2\pi}{T}\text{rad/s}$

8. $\omega = \frac{2\pi}{T}\text{rad/s}$

9. $\theta = \omega t + \frac{1}{2}\alpha t^2$

10. $w_i = 0$

$N = 50t = 2\text{sec}$

$\theta = 2\pi N = 2\pi \times 50 = 100\pi\text{rad}$ $\theta = \frac{1}{2} \propto t^2$

$\alpha = \frac{2\theta}{t^2} = 50\pi\text{rad/s}^2$ $'\theta\ '$ in 4sec

$\theta = \frac{1}{2}\alpha t^2 = \frac{1}{2} \times 20\pi \times 16 = 400\pi$

in last 2sec $\theta = 400\pi - 100\pi = 300\pi$

$N = \frac{\theta}{2\pi}$ $N = 150$

11. $\omega_1 = 0$, $\omega f = 100\text{rev/sec.}$, $t = 4\text{sec}$

$\alpha = \frac{\omega f - \omega i}{t} = \frac{100 - 0}{4} = 25 rev/sec^2$

12. $\theta = \omega_1 t + \frac{1}{2}\alpha t^2$

$\theta = 0.(4) + \frac{1}{2}(25 \times 2\pi)(4)^2$

$\theta = \frac{1}{2}(50\pi)(16)$

$\theta = 400\pi rad.$

13. $\omega_1 = 2 rad/s,\ \alpha = 3 rad/s^2,\ t = 2sec$

$\theta = \omega_1 t + \frac{1}{2}\alpha t^2$

$\theta = (2)(2) + \frac{1}{2}(3)(2)^2$

$= 4 + 3(2)$

$\theta = 10 rad.$

14. $N = 50 \rightarrow t_1 = 5sec$

$\alpha = ?\ \omega = ?$

$\theta = 2\pi N = 2\pi \times 50 = 100\pi rad$

$\theta = \frac{1}{2}\alpha t^2,\ \alpha = \frac{2\theta}{t^2} = \frac{200\pi}{25} = 8\pi rad/s^2$

$\alpha = 8\pi rad/s^2,\ \omega^2 = 2\alpha\theta$

$\omega^2 = 2 \times 100\pi \times 8\pi,\ \omega^2 = 1600\pi^2$

$\omega = 40\pi rad/sec$

15. $f = 6rpm$

$= \frac{6}{60} rps$

$= \frac{1}{10} rps$

$\omega = 2\pi f = 2\pi \cdot \frac{1}{10}$

16. $\theta = \omega t = \frac{2\pi}{60} \times 45 = \frac{3\pi}{2} rad$

17. $\omega_2^2 - \omega_1^2 = 2\alpha\theta$

18. $\omega_1 = 5 rad/s,\ \omega_2 = 15 rad/s$

$\alpha = 1 rad/s^2$

$\omega_2^2 - \omega_1^2 = 2\alpha\theta$

$(15)^2 - (5)^2 = 2(1)\theta$

$225 - 25 = 2.\theta$

$200 = 2\theta$

$\theta = 100 rad.$

18. $\omega = \frac{2\pi}{60} = \frac{\pi}{30}$

$\omega = 0.1047 rad/s$

$V = rw$

$V = 3 \times 10^{-2} \times 1.047$

$V = 0.00314$

19. B

20. $\theta_1 = \frac{1}{2}\alpha t^2 = \frac{1}{2}(4)(100) = 200 rad$

$\omega_2 = \omega_1 + \alpha t = 0 + 4(10) = 40\text{rad/s}$

$\theta_2 = \omega_2 t = 40(10) = 400\text{rad}$

$\theta_3 = \omega_2 t - \frac{1}{2}\alpha t^2 = 40(10) - \frac{1}{2}(4)(100) = 200\text{rad}$

$\therefore \theta = \theta_1 + \theta_2 + \theta_3 = 200 + 400 + 200 = 800\text{rad}$

LEVEL - II

21. A-R, B-Q, C-P, D-S

22. A-R, B-Q, C-S, D-P

23. A, B, C, D

24. $\theta = 6\text{t} + 3\text{t}^2 + 4\text{t}^3$

$\frac{d\theta}{df} = 6 + 6t + 12t^2$

$\frac{\text{d}^2\theta}{\text{dt}^2} = 6 + 24\text{t}$

$\alpha = 6 + 24\text{t}$

$\text{t} = 2\text{sec} \Rightarrow \alpha = 6 + 24(2)$

$= 6 + 48$

$= 54\text{rad/sec}^2$

25. $\omega = 6 + 6t + 12t^2$

$\text{t} = 1\text{sec}$

$\omega = 6 + 6(1) + 12(1)$

$= 6 + 6 + 12$

$\omega = 24\text{rad/s}$

26.

$\omega = 6 + 6t + 12t^2$

$\text{t} = 0 \Rightarrow \text{w} = 6\text{rad/s}$

27. $\omega_1 = \frac{600(2\pi)}{60}$

$\omega_1 = 10(2\pi)^{\text{rad/s}}$

$\omega_2 = \frac{1200(2\pi)}{60}$

$\omega_2 = 20(2\pi)\text{rad/1}$

$\text{t} = 5\text{sec}$

$\alpha = \frac{\omega_2 - \omega_1}{\text{t}} = \frac{(20-10)(2\pi)}{5}$

$= \frac{10.2\pi}{5}$

$\alpha = 4\pi \cdot rd/s^2$

28. $\omega^2 - \theta^2 = 2\alpha\theta$

$\alpha = \frac{\omega^2}{2\theta} = (2\omega)^2 - \omega^2 = 2\alpha \cdot \theta^1$

$= 3\omega^2 = 2 \cdot \frac{\omega^2}{2\theta} \cdot \theta^1 \Rightarrow \theta^1 = 3\theta$

29. $\text{D} = 3\text{ m};\ r = \frac{3}{20} = 1.5\text{ m}$

$\text{V} = 18\text{ m/s}$

$\text{V} = \text{rw}$

30. $r = 25 \times 10^{-2}$ m

$V = 5$ m/s

$\omega = \frac{v}{r}$

$\omega = \frac{5}{25 \times 10^{-2}} = \frac{500}{25} = 20.\text{rad/s}$

31. $\omega_1 = 10,\ \omega_2 = 60\text{rad/s},\ t = 30\text{sec}$

$\omega_2 = \omega_1 t\alpha t$

$60 = 0 + \alpha(30)$

$\alpha = 2\text{rad/s}^2$

32. $\omega_1 = 10(2\pi)^{\text{rad}}/\text{s},\ t = 14\text{sec}, \omega_2 = 0$

$\omega_1 = 10(2\pi) \cdot \text{rad/s}$

$\omega_2 = 0,\ t = 14\text{sec}$

$\omega_2 = \omega_1 + \alpha.t$

$0 = 20\pi + \alpha.$

$\alpha = \frac{-20\pi}{14}$

$\omega_2^2 - \omega_1^2 = 2\alpha\theta$

$\Rightarrow 0^2 = (20\pi)^2 = 2\left(\frac{20\pi}{14}\right)\theta$

$\Rightarrow 20\pi = \frac{1}{7}\theta$

$10(2\pi) = \frac{1}{7}\theta \Rightarrow 70(2\pi) = \theta \Rightarrow 70$ rer.

WORKSHEET – 2

LEVEL - I

1. D	2. D	3.3 C	4. D	5. B	6. A	7. B	8. C
9. D	10. D	11. D	12. A	13. B	14. D	15. B	

LEVEL – II

1. D	2. B	3. A	4. D	5. C	6. C	7. A	8. D
9. D	10. C	11. A	12. 2	13. B	14. C	15. D	16. A
17. D	18. B	19. C					

LEVEL – III

20. A, B, C	21. A	22. B	23. B	24. B	25. B
26. a-r, b-p, c-s, d-q		27. B	28. B	29. A	30. A

SOLUTIONS

LEVEL – II

1. $\vec{\tau} = \vec{r} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 2 & 0 \\ 2 & -3 & 0 \end{vmatrix} = \hat{i}(0) - \hat{j}(0) + \hat{k}(-9 - 4) = -13\hat{k}$

2. $\tau = rF\sin\theta = 40 \times 10^{-2}(5)\sin 30^\circ = 1\text{Nm}$

3. $\tau = r \times F$

$= 20 \times 10^{-2} \times 8 = 160 \times 10^{-2}$

$\tau = 1.6\text{NM}$

4. $\text{F} = 10\ \text{N}\ \tau = 5\text{Nm}$

$\vec{\text{r}} = ?\quad \vec{\tau} = \vec{\text{r}} \times \vec{\text{F}}$

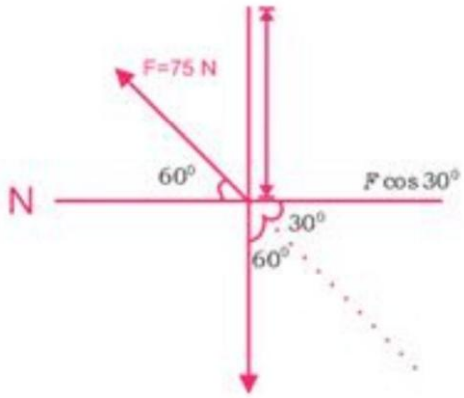
DE3HY $\vec{\text{r}} = \frac{\text{T}}{\text{F}}\quad = \frac{5}{10} = 0.5\ \text{m}$

5. $\text{F} = 10\ \text{N}$

$r = 50\ \text{cm} = 0.5\ \text{m}$

$\tau = \text{rF}\ = 0.5 \times 10\ \tau = 5\text{Nm}$

6.



$\tau = r \times F_2 \quad r = 0.8m$

$F = 75N \quad \tau = 0.8 \times 75\cos\ 30° = 0.8 \times 75 = 30N - m$

7. $\text{F} = 120\ \text{N}, \text{r} = \text{I} = 40\ \text{cm} = 0.4\ \text{m}$

$\tau = \text{I} \times \text{F} = 0.4 \times 120\ \tau = 48\text{Nm}$

$\tau = 48\text{Nm}, \text{F} = 40\ \text{N}\ \text{r} = ?\ \tau = \text{r} \times \text{Fr} = \frac{\tau}{\text{F}} = \frac{48}{40} = \frac{6}{5}$

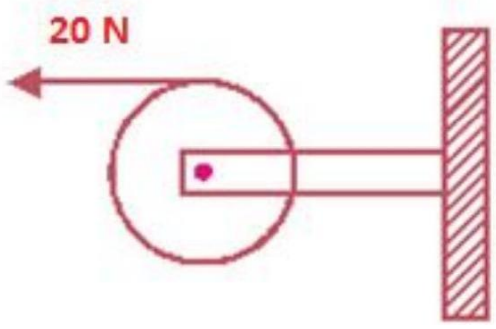
8. $\vec{F} = (\hat{\imath} + 2j\hat{\jmath} - 3\hat{k}), \vec{r} = (2\hat{\imath} + 3\hat{\jmath} - \hat{k})$

$\vec{\tau} = \vec{r} \times \vec{F}\ = (2\hat{\imath} + 3\hat{\jmath} - \hat{k}) \times (\hat{\imath} + 2\hat{\jmath} - 3\hat{k})$

$= \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 2 & 3 & -1 \\ 1 & 2 & -3 \end{vmatrix}$

$\tau = \hat{\imath}(-9 + 2) + j(-1 + 6) + k(4 - 3) = (-7\hat{\imath} + 5\hat{\jmath} + \hat{k})N - m$

9.



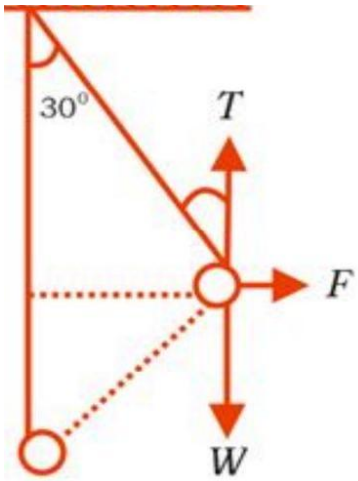
$\text{r} = 20\ \text{cm}\ = 0.2\ \text{m}\ \ \text{F} = 20\ \text{N}$

$\tau = r \times \vec{F} = 0.2 \times 20 = 4\text{Nm}$

10. $\tau = rF \Rightarrow 0.5 = r(50)$

$\text{r} = \frac{5 \times 10^{-1}}{50} = 1\ \text{cm}$

12.

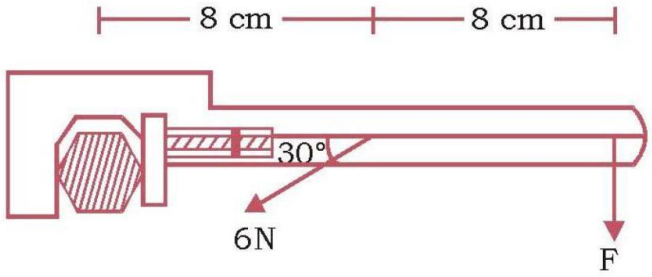


$\tau = rF_{\perp}\ F_1 = W\cos\ 60°$

$\tau = I \times F_{\perp} = 1 \times 1 \times 10 \times \frac{1}{2}$

$\tau = 5\ N.m$

13. $\tau = w\ell\sin\ \theta$ (Conceptual)



$\tau = r_{\perp} \cdot F\ \tau = r.F_{\perp}$

$= 8\sin\ 30° \times 6 \quad 8 \times 6\sin\ 30°$

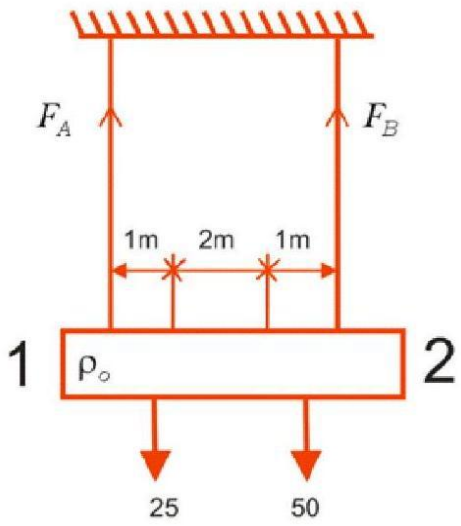
$= 0.24Nm\ = 0.24Nm$

$= 0.24Nm\ \tau = \frac{15}{100} \times F$

$0.24 = \frac{16}{100} \times F\ F = \frac{24}{16} = 1.5\ N$

15. $\vec{\tau} = \vec{r} \times \vec{F} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 2 & 1 & 0 \\ 1 & 2 & 0 \end{vmatrix} = \hat{\imath}(0) - \hat{\jmath}(0) + \hat{k}(3) = 3\hat{k}$

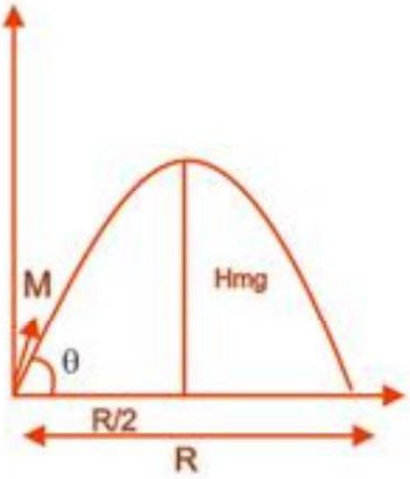
16.



Taking moment of forces
Torque about P,

$$T_B \times 4 = 50 \times 3 + 25 \times 1\ 4\ T_B = 175\ \ T_B = \frac{175}{4} = 43.75\ N$$

17.



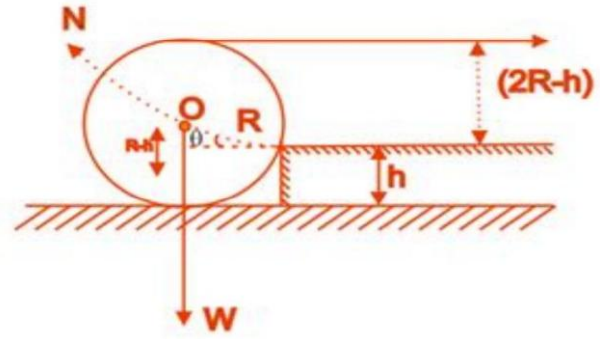
$$\vec{\tau} = \vec{F} \times \vec{r},\ \vec{\tau} = mg(R/2) = \frac{mgu^2 \sin\ 2\theta}{2g},\ \tau = \frac{1}{2}mu^2 \sin\ 2\theta$$

18. $\bar{r} = (2\hat{\imath} - 3\hat{\jmath} + 6\hat{k}) - (2\hat{\imath} - 5\hat{\jmath} + 3\hat{k}) = 2\hat{\imath} + 3\hat{k}$

$$\vec{\tau} = \vec{r} \times \vec{F} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 0 & 2 & 3 \\ 2 & 3 & -4 \end{vmatrix} = \hat{\imath}(-8-9) - \hat{\jmath}(0-6) + \hat{k}(-4)$$

$$= (-17\hat{\imath} + 6\hat{\jmath} - 4\hat{k})\text{Nm}$$

19.



$d^2 = R^2 - (R-h)^2$ $d = \sqrt{R^2-(R-h)^2}$

$d = \sqrt{2Rh - h^2}$ Taking torque about $P, \tau_F > \tau_w$

For minimum F, $F(2R-h) = w \times d$, $F = \frac{W\sqrt{2Rh-h^2}}{(2R-h)}$

LEVEL – III

20. A, B, C

21. $\tau = \mathrm{r} \times \mathrm{F}$

$\tau = (-i - 2j) \times (xi + yj)$

$= -\mathrm{y}(\mathrm{i} \times \mathrm{j}) - 2\mathrm{x}(\mathrm{j} \times \mathrm{i}) = -\mathrm{yk} + 2\mathrm{xk}$

$\tau = (2x - y)\hat{k} = [2(2) - (-3)]\hat{k} = (4+3)\hat{k}$

$\tau = 7\hat{\mathrm{k}}$

22. $\tau = (2\mathrm{x} - \mathrm{y})\hat{\mathrm{k}}$

$= [2(2) - (-3)]\hat{\mathrm{k}} = (-4+3)\hat{\mathrm{k}}$

$\tau = -\hat{k}$

23. The net moment acting on the rod = Sum of all the torque acting unit

$\tau_{\text{net}} = \tau_1 + \tau_2 = 17 + 25 = 42\text{Nm}$

24. $\frac{\mathrm{F_1}}{\mathrm{F_2}} = \frac{\mathrm{d_2}}{\mathrm{d_1}}$; $\frac{10\times10}{\mathrm{F_2}} = \frac{1}{1}$

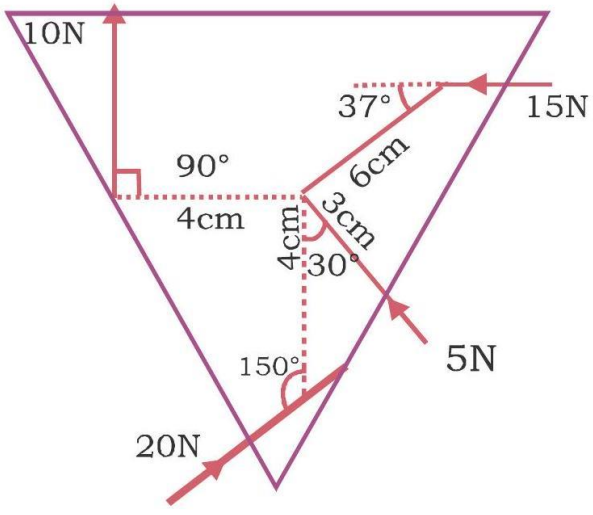
$\mathrm{F_2} = 10 \times 10;\ \mathrm{m_2} \times 10 = 100$

$\therefore \mathrm{m_2} = 10\ \mathrm{kg}$

25. $160 \times 30 = 240 \times x$

$\mathrm{x} = 20\ \mathrm{cm}$

26.



Torque about a point = Total force × perpendicular from the point to 15 at force let Torque in cW(-ve) and in A. cW (+1)

a) Torque of 5 N force about. The point ' O ' is $= 5 \times 3 \times 10^{-2} \times \sin\theta =$ 0 N − m

b) torque of 10 N force about the, point

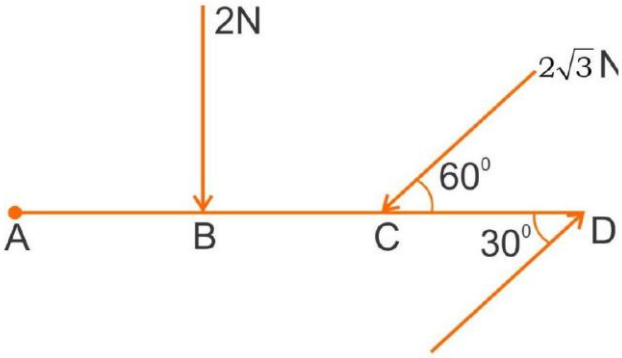
'O' is $= 10 \times 4 \times 10^{-2} \cdot \sin 90° = 4 \times 10^{-2}$ (clockwise)

c) Torque of 15 N force about ' O ' is $= 15 \times 6 \times 10^{-2} \cdot \sin 37°$

$= +54 \times 10^{-2}$(A.C.W)

d) Torque of 20 N force about ' O ' is $= 20 \times 4 \times 10^{-2} \times \sin 30°$

$= +40 \times 10^{-2}$(A.C.W)



6 N

27. Torque due to 2 N force is $= 1 \times 2\sin 90°$
$= 1 \times 2\sin 90°$
$= -2$ N − m(CW

28. Torque due to $2\sqrt{3}N$ force is
$= 2 \times 2\sqrt{3}\sin 60° = -6$Nm(CW)

29. Torque to 6 N force is
$= 3 \times 6\sin 30°$
= 9Nm(A.C.W)

30. Net torque about hinge ' A ' is
$= -2 - 6 + 9 = 1$Nm

WORKSHEET – 3

LEVEL – I

1. D	2. A	3. B	4. B	5. D	6. A	7. A	8. A
9. B	10. B	11. A	12. C	13. D	15. C		

LEVEL – II

1. D	2. A	3. D	4. D	5. C	6. B	7. B	8. B
9. C	10. B	11.B	12. A	13. D	14. A	15. D	16. C
17. A	18. A, B, C, D						

LEVEL – III

18. A, B, C, D
19. a-p, b-s, c-r, d-q
20. a-q, b-s, c-s, d-p

21. A 22. B 23. A 24. C 25. D 26. D 27. C 28. C
29. D 30. A 31. B

SOLUTIONS
LEVEL – II

1. $\text{mass} = \rho \times V$

$m_1 = \rho \times \frac{4}{3}\pi R_1^2;\ m_2 = \rho \times \frac{4}{3}\pi R_2^2$

$\frac{I_1}{I_2} = \frac{\frac{2}{5}m_1R_1^2}{\frac{2}{5}m_2R_2^2};\ \frac{I_1}{I_2} = \frac{\rho \times \frac{4}{3}\pi R_1^2 \times R_1^2}{\rho \times \frac{4}{3}\pi R_2^2 \times R_2^2} = \frac{R_1^4}{R_2^4}$

$= \left(\frac{R_1}{R_2}\right)^4 = \left(\frac{R_1}{2R_1}\right)^4 = \frac{1}{16};\ \frac{I_1}{I_2} = \frac{1}{16}$

$I_2 = 16I_1$

2. $m_1r_1^2 + m_2r_2^2$

3. $I = m_1r_1^2 + m_2r_2^2 + m_3r_3^2; I = 1 \times (1)^2 + 2(2)^2 + 3(3)^2$

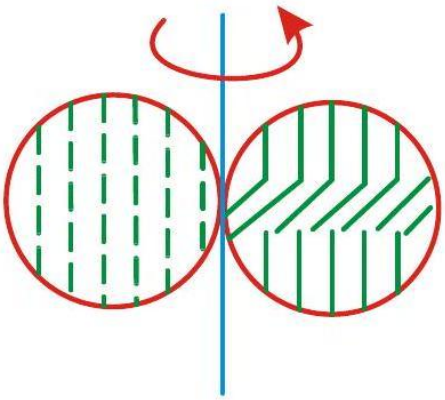
$I = 1 + 8 + 27;\ I = 36\ \text{g cm}^2$

$R = \sqrt{\frac{I}{m}} = \sqrt{\frac{36}{6}} = \sqrt{6} = 2.449\ \text{cm}$

4. $I = \frac{ML^2}{12}, I_z = I_x + I_y;\ \therefore I^1 = \frac{ML^2}{12} + \frac{ML^2}{12} = \frac{ML^2}{6};\ \therefore I^1 = 2I$

5. $\Rightarrow k = \sqrt{\frac{1}{m}} = \sqrt{\frac{0.5}{2}} = 0.5\ \text{m} - 50\ \text{cm}$

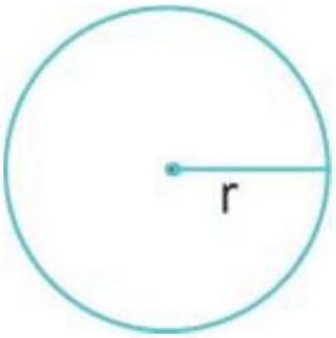
6. $\frac{I_{\text{centre}}}{I_{\text{tan gent}}} = \frac{\frac{2}{5}MR^2}{\left(\frac{2}{5}MR^2 + MR^2\right)} = \frac{2}{7}$



$I_1 = 2I \& I = \frac{5}{4}MR^2$

$\Rightarrow I = 2 \times \frac{5}{4}mR^2 = \frac{5}{2}mR^2 = 2.5 \times 0.1 \times \left(10 \times 10^{-2^2}\right) = 2.5 \times 10^{-3}\text{kgm}^2$

8.

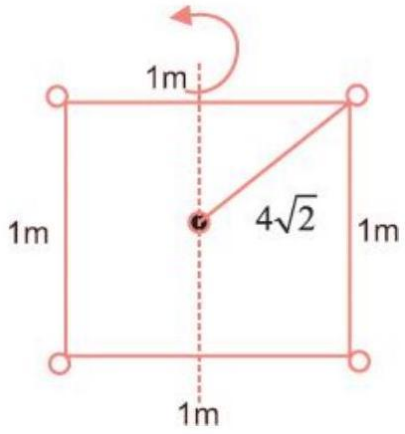


$$2\pi r = l;\ r = \frac{l}{2\pi}$$

Moment of Inertia = mr^2

$$I = m\left(\frac{l}{2\pi}\right)^2;\ I = \frac{ml^2}{4\pi^2}$$

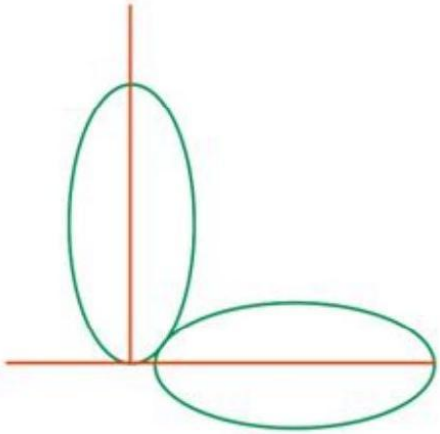
9.



$$I = m_1r_1^2 + m_2r_2^2 + m_3r_3^2 + m_4r_4^2 = 4\ m\left(\frac{l}{\sqrt{2}}\right)^2 = 4\ m\left(\frac{l}{\sqrt{2}}\right)^2 = \frac{4\ m}{2} = 2\ ml^2$$

$$I = MK^2;\ 2\ m = 4mK^2;\ \frac{2}{4} = K^2;\ K = \frac{1}{\sqrt{2}}$$

10. $I = 2I_{\text{tan gent}}$ and $I_{\text{tan gent}} = \frac{7}{5}MR^2; \therefore I = 2 \times \frac{7MR^2}{5} = \frac{14MR^2}{5}$



$$I = 2MR^2 + \frac{MR^2}{2};\ I = \frac{5MR^2}{2}$$

12. Since these cases are same as circular disc. The distributions with respect to axis of rotation is same pattern as circular disc

$\therefore I = \frac{MR^2}{2}$

13. $\frac{R_1}{R_2} = \frac{1}{2}$ $m_1 = D_1 V_1$; $m_1 = D \times \frac{4}{3}\pi R_1^3$; $m_2 = D \times \frac{4}{3}\pi R_2^3$;

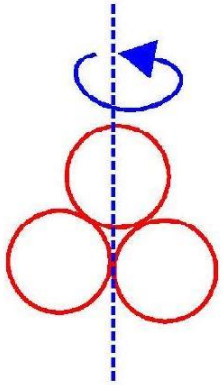
$\frac{I_1}{I_2} = \frac{\frac{2}{5}m_1R_1^2}{\frac{2}{5}m_2R_2^2} = \frac{D \times \frac{4}{3}\pi R_1^3 R_1^2}{D \times \frac{4}{3}\pi R_2^3 R_2^2}; \frac{I_1}{I_2} = \frac{R_1^5}{R_2^5} = \left(\frac{1}{2}\right)^5 = \frac{1}{32}$

14. $I_{ring} = I_{disc}$;

$r_{ring} = r_{disc}$

$\frac{I_{ring}}{I_{disc}} = \frac{M_{ring}\, r_{ring}^2}{\frac{M_{disc}\, r_{disc}^2}{2}}; \frac{M_{ring}}{M_{disc}} = \frac{1}{2}$

15.



$\therefore I = \frac{MR^2}{2} + 2\left[\frac{3}{2}MR^2\right] = \frac{7MR^2}{2}$

16. With respect to the Centre of mass effective mass $= \frac{m_1m_2}{m_1+m_2}$

$\therefore I = \left(\frac{m_1m_2}{m_1+m_2}\right)d^2$

17. $I = \frac{MR^2}{2}$

$\frac{I_1}{I_2} = \frac{\frac{MR_1^2}{2}}{\frac{MR_2^2}{2}} = \frac{R_1^2}{R_2^2} = \frac{\rho_2}{\rho_1} = \frac{8.9}{7.2}$

LEVEL – III

18. A, B, C, D
19. a-p, b-s, c-r, d-q
20. a-q, b-s, c-s, d-p
21. $I_{earth} = \frac{2}{5}mR^2 = \frac{2}{5} \times 10^{25} \times (6.4 \times 10^6)^2 = 16.384 \times 10^{37} = 1.6384 \times 10^{38}$ kg m^2
22. $I = 4m\left(\frac{l}{\sqrt{2}}\right)^2$

$I = 2mll^2 = 2$ kg m^2

23. $I = \frac{mR^2}{12}$

$k = \frac{l}{2\sqrt{3}} = \frac{150}{2\sqrt{3}} = 43.3$ cm

24. $I = I_1 + I_2 + I_3 + I_4$

$= M\left(\frac{L}{\sqrt{2}}\right)^2 + M\left(\frac{L}{\sqrt{2}}\right)^2 + M\left(\frac{L}{\sqrt{2}}\right)^2 + M\left(\frac{L}{\sqrt{2}}\right)^2$

$= 4M\frac{L^2}{2} = 2ML^2$

25. $I = MK^2$ and $I = 2ML^2 \Rightarrow 2ML^2 = MK^2$

$K^2 = 2\ L^2$

$K = \sqrt{2}\ L$

26. $I = ml^2 + ml^2 + m(\sqrt{2}l)^2;\ I = 4ml^2$

27. $I = 2\left[m\left(\frac{l}{\sqrt{2}}\right)^2\right];\ I = ml^2$

28. $I = ml^2 + ml^2;\ I = 2ml^2$

29. $I = \frac{mb^2}{3}$

30. $I = \frac{ml^2}{12}$

31. $I = \frac{mb^2}{12}$

WORKSHEET – 4

LEVEL – I

1. B	2. A	3. A	4. B	5. B	6. C	7. B	8. A
9. B	10. A						

LEVEL – II

1. C	2. D	3. C	4. B	5. C	6. B	7. A	8. B
9. D	10. B	11. D	12. C	13. A	14. A	15. A	16. B
17. C	18. C						

LEVEL – III

19. a-r, b-s, c-p, d-q	20. B	21. A	22. A	23. A
24. C	25. A	26. A	27. D	28. A	29. A

SOLUTIONS

LEVEL – I

1. A circular disc is rotating about its own axis the direction of its angular momentum is
   Along axis of rotation
2. Since the ice that was at the South Pole is now distributed around the world, there is a change in the distribution of mass. Like an ice skater opening their arms. Since the mass is further from the axis of rotation, the rotation slows; which means the days are longer.

3. If most of the population on earth is migrated to poles of earth then the time duration of a day Increases
4. If polar ice caps melts then the angular velocity of rotation of earth Decreases
5. The law of conservation of angular momentum is obtained from newton's II law in rotational motion when External torque is zero
6. If radius of earth shrinks then the duration of day Decreases
7. A diver folds himself close to his axis of rotation while diving down Angular momentum of diver remains constant

LEVEL – II

1. $v = r\omega = \frac{r(2\pi)}{T}$

$\hat{L} = \hat{r} \times \hat{p} \Rightarrow L = mvr = \frac{m(2\pi r)}{T} r = \frac{2\pi r^2 m}{T} (\theta = 90°)$

2. Given n = 100rpm $= \frac{100}{60}$rps

From LCAM, $I_1\omega_1 = I_2\omega_2$

$I\ (2\pi)\left(\frac{100}{60}\right) = \left(\frac{1}{2}\right) 2\pi n$

$n = \frac{100}{60}(2)\text{rps} = 200\text{rpm}$

3. $I_1\omega_1 = I_2\omega_2$

$\omega_1 = \omega, \omega_2 = 2\omega$

$\frac{I_1}{I_2} = \frac{\omega_2}{\omega_1} = \frac{2\omega}{\omega} = \frac{2}{1}$

$I_1 : I_2 = 2:1$

4. $L = I\omega = \frac{mr^2}{2}\omega$

5. $\frac{r_1}{r_2} = \frac{2}{3} : \frac{T_1}{T_2} = \frac{2}{3} \Rightarrow \frac{\omega_1}{\omega_2} = \frac{3}{2}$

$\frac{L_1}{L_2} = \frac{m_1 v_1 r_1}{m_2 v_2 r_2} = \frac{m_1 r_1^2 \omega_1}{m_2 r_2^2 \omega_2} = \frac{m}{2m}\left(\frac{4}{9}\right)\left(\frac{3}{2}\right) = \left(\frac{1}{3}\right)$

6. $I_1\omega_1 = I_2\omega_2$

$I_1 = I, I_2 = \frac{3}{4}I$

$\omega_1 = 180\text{rev/min}$

$= \frac{180}{60} = 3\text{rev/min}$

$\omega_2 = \frac{I_1}{I_2} \cdot \omega_1$

$= \frac{1}{\frac{3}{4}I} \cdot (3)$

$\omega_2 = \frac{4}{3} \cdot 3$

$\omega_2 = 4\text{rev/sec}$

7. $I_1\omega_1 = I_2\omega_2$

$Mr^2\omega = (Mr^2 + 2mr^2)\omega_2$

$\omega_2 = \dfrac{M\omega}{M+2m}$

8. $\mathrm{I} = \frac{\mathrm{mL}}{3} \cdot \omega$

$= \dfrac{7\times10^{-3}\times10^{-3}\times(1.5\times10^{-2})^2}{3}\times\dfrac{\pi}{30}$

$= \dfrac{7\times10^{-6}\times2.25\times10^{-4}}{3}\times0.03$

$I = 5.5\times10^{-11}\mathrm{Kgm^2\ s^{-1}}$

9. $\mathrm{L} = \mathrm{I}\omega$

$= \dfrac{\mathrm{mR}^2}{2}\cdot\omega$

$= \dfrac{4\times(10\times10^{-2})^2}{2}\times5$

$= 10\times100\times10^{-4}$

$\mathrm{L} = 0.1\mathrm{Kgm^2\ s^{-1}}$

10. $\mathrm{L} = \mathrm{I}\omega = \text{constan t}$

$r^2\omega = \text{cons tan t}$

$\omega\infty\dfrac{1}{\mathrm{r}^2} \Rightarrow \dfrac{\Delta\omega}{\omega}\times100 = -2\left(\dfrac{\Delta\mathrm{r}}{\mathrm{r}}\times100\right)$

$\dfrac{\Delta\omega}{\omega}\times100 = -2(0.2)$

i.e., decreases by 0.4%

11. $\mathrm{I}_1\omega_1 = \mathrm{I}_2\omega_2 \Rightarrow \frac{2}{5}\mathrm{mr}^2\left(\frac{2\pi}{\mathrm{T}_1}\right) = \frac{2}{5}\ \mathrm{m}\left(\frac{\mathrm{r}}{2}\right)^2\left(\frac{2\pi}{\mathrm{T}_2}\right)$

$\dfrac{\mathrm{r}^2}{\mathrm{T}_1} = \dfrac{\mathrm{r}^2}{4}\left(\dfrac{1}{\mathrm{T}_2}\right)$

$\dfrac{1}{24} = \dfrac{1}{4\ \mathrm{T}_2}$

$\mathrm{T}_2 = 6\mathrm{hrs}$

12. $r_1 = r, r_2 = \frac{3r}{4}, I_1\omega_1 = I_2\omega_2\ \left(\omega = \frac{2\pi}{T}\right)$

$\dfrac{2}{5}mr^2\left(\dfrac{2\pi}{24}\right) = \dfrac{2}{5}m\dfrac{9r^2}{16}\left(\dfrac{2\pi}{T_2}\right)$

$24 = \dfrac{16\ \mathrm{T}_2}{9} \Rightarrow \mathrm{T}_2 = 13.5\mathrm{hrs}$

13. A

14. A

15. $\tau = l\alpha$

$100 = I(2)$

$\mathrm{I} = 50\mathrm{Kgm^2}$

16. $\tau = I_{\text{diametre}}\cdot\alpha$

$\alpha = \dfrac{2\omega-\omega}{3} = \dfrac{\omega}{3}$

$\mathrm{I}_{\text{diametre}} = \dfrac{2\mathrm{mR}^2}{5}$

$\tau = \dfrac{2mR^2}{5}\times\dfrac{\omega}{3}$

$\tau = \frac{2mR^2}{15}$

17. $\bar{L} = \bar{r} \times \bar{P}$

$\bar{L} = (2\hat{i} + 3\hat{j} + 4\hat{k}) \times (3\hat{i} + 4\hat{j} + 5\hat{k})$

$L = (2i + 3j + 4k) \times (3i + 4j + 5k)$

$\begin{vmatrix} i & j & k \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix}$

$= i(15 - 16) - j(10 - 12) + k(8 - 9)$

$= -i + 2j - k$

18. $\tau = I_{edge} \cdot \alpha$

$\alpha = \frac{4\omega - 2\omega}{3} = \frac{2\omega}{3}$

$\tau = \frac{mL^2}{3} \cdot \frac{2\omega}{3}$

$\tau = \frac{2\ mL^2}{9}$

LEVEL – III

19. a-r, b-s, c-p, d-q

20. $K = \frac{1}{2}(2)(2)$

$K = 2\ J$

21. $K = \frac{1}{2} I\ \omega^2$

$= \frac{1}{2}\frac{2}{5} mR^2 \cdot \omega^2$

$= \frac{1}{2}\frac{2}{5}(2)(1)^2 \cdot (4)^2$

$= \frac{32}{5} = 6.4\ J$

22. $\tau = 1\alpha$

$\alpha = \frac{5\omega - 2\omega}{3} = \omega$

$\tau = I\omega$

23. $\tau = I\alpha$

$\alpha = \frac{4\omega - 2\omega}{3} = \frac{2\omega}{3}$

$\tau = \frac{m\ell^2}{12} \cdot \frac{2\omega}{3}$

$\tau = \frac{ML^2\omega}{18}$

24. $\alpha = \frac{\tau}{I}$

$= \frac{40}{2}$

$\alpha = 20 rad/sec^2$

$\omega_1 = 0, \omega_2 = 100 rad/s$

$\Rightarrow \omega_2 = \omega_1 + \alpha t$

$100 = 0 + 20.t$

$t = 5sec$

25. $\tau = \frac{dL}{dt} = \frac{\frac{3L}{4} - L}{2}$

$\tau = \frac{L}{8}$

26. $K_1 = \frac{L^2}{21} = E$

$K_1 = E, L_1 = L, L_2 = 2\ L$

$I_2 = \frac{I}{2}$

$I_1 = I$

$K_2 = \frac{L_2^2}{2I_2}$

$K_2 = \frac{(2\ L)^2}{2\left(\frac{I}{2}\right)}$

$= 8\frac{L^2}{2I}$

$K_2 = 8E$

27. Conceptual

28. $v_1 = \frac{v_2}{64} \Rightarrow R_1 = \frac{R_2}{4}, n = 4$

$T_2 = \frac{T_1}{n^2} = \frac{24}{16} = 1.5hrs$

29. $K = \frac{L^2}{21}, r_1 : r_2 = 2:1$

$K_1 = K_2, I_1 = \frac{2}{5}mr_1^2$

$I_1 = \frac{2}{5}\ m(2)^2$

$I_2 = \frac{2}{5}\ m(1)^2$

$\frac{L_1^2}{2I_1} = \frac{L_2^2}{2I_2}$

$\frac{L_1}{L_2} = \sqrt{\frac{I_1}{I_2}}$

$\frac{L_1}{L_2} = \frac{2}{1} = 2:1$

WORKSHEET – 5
LEVEL – I

1. A 2. C 3. D 4. D 5. A 6. B 7. B 8. A 9. D

LEVEL – II

1. A	2. C	3. B	4. B	5. D	6. C	7. A	8. B
9. C	10. A	11. A	12. A	13. A	14. B	15. D	16. C
17. C	18. A						

LEVEL – III

19. a-r, b-s, c-p, d-q

20. A, C, D	21. B	22. D	23. D	24. D	25. C	26. D
27. A						

SOLUTIONS

LEVEL – II

1. $\tau = I\,\alpha$

$1000 = 200\left(\frac{\omega_2 - \omega_1}{t}\right)$

$5 = \left(\frac{\omega_2 - 0}{3}\right)$ $[\because \omega_1 = 0]$

$\omega_2 = 15\text{rad/s}$

2. $W = 424$ J

$\omega_1 = 60\text{rpm} = \frac{60}{60} \times 2\pi$

$\omega_2 = 360\text{rpm} = \frac{360}{6} \times 2\pi$

$= 2\pi\text{rad/s}$

$W = K_2 - K_1$

$= \frac{1}{2} I(\omega_2^2 - \omega_1^2)$

$484 = \frac{1}{2} I((12\pi)^2 - (2\pi)^2)$

$484 = \frac{1}{2} I(140\pi^2)$

$I = 0.7\text{Kgm}^2$

3. $P = \tau\omega$

$3.14 \times 10^3 = \tau\left(\frac{3000}{60}\right)$

$\tau = 10\text{Nm}$

4. $\alpha = \frac{\omega_2 - \omega_1}{t} = \frac{0-20}{4} = -5\text{rad/s}^2$

$\tau = I\alpha = 0.2 \times 5 = 1\text{Nm}$

$\omega_2 = \omega_1 + \alpha t = 20 + (-5)2 = 10\text{rad/s}$

$\text{Workdone} = \frac{1}{2} I(\omega_2^2 - \omega_1^2)$

$= \frac{1}{2} 0.2(100 - 400) = -30$ J

5. $\tau = I\alpha$

$\tau = (6 \times 10^{-2})\left(\frac{20}{4}\right)$

$= 30 \times 10^{-2}$

$\tau = 0.3\text{Nm}$

6. $\mathrm{K} = \frac{1}{2}\mathrm{I}\omega^2$

$$62.8 = \frac{1}{2}I\left(\frac{300 \times 2\pi}{60}\right)^2$$

$$62.8 = \frac{1}{2}\mathrm{I}(31.4)^2$$

$$I = 0.12739$$

$$\mathrm{L} = \mathrm{I}\omega$$

$$= (0.12739)(31.4)$$

$$\mathrm{L} = 4 \text{ J S}$$

7. $\Delta\mathrm{K} = \frac{1}{2}(\Delta\mathrm{L})(\Delta\omega)$

$$\Delta\mathrm{K} = \frac{1}{2}(60)(30 - 10)$$

$$\Delta\mathrm{K} = \frac{1}{2}(60)(20)$$

$$\Delta\mathrm{K} = 600 \text{ J}$$

8. $\alpha = \frac{0-10}{4} = -2.5\text{rad/s}^2$

$$\tau = 1\alpha = (0.2)(2.5) = 0.5\text{Nm}$$

$$\omega_2 = 10 + (-2.5)2 = 5$$

$$w = \frac{1}{2}\mathrm{I}(\omega_2^2 - \omega_1^2)$$

$$= \frac{1}{2}\mathrm{I}(0.2)(25 - 100)$$

$$\mathrm{w} = -7.5 \text{ J}$$

$$|\mathrm{w}| = 7.5 \text{ J}$$

9. $p = \tau\omega$

$$p = (100)\left(\frac{1260}{60} \times 2\pi\right)$$

$$\mathrm{p} = 13.2\mathrm{k}\omega$$

10. $\omega = 2\pi n$

$$\text{Work done} = \frac{1}{2}I(4\pi^2 n_2^2 - 4\pi^2 n_1^2)$$

11. $\mathrm{K.E} = \frac{1}{2}I(\omega_2^2 - \omega_1^2)$

$$1500 = \frac{1}{2}(1.2)(\omega_2^2 - 0)$$

$$\omega_2 = 50\text{rad/s}$$

$$\alpha = \frac{\omega_2 - \omega_1}{t} \Rightarrow \mathrm{t} = \frac{50 - 0}{25} = 2\text{sec}$$

12. $\mathrm{P} = \frac{\mathrm{w}}{\mathrm{t}} = \frac{\frac{1}{2}\mathrm{I}\omega^2}{\mathrm{t}} \Rightarrow \omega = 60\pi$

$$\theta = 2\pi\mathrm{n} \Rightarrow \mathrm{n} = \frac{\theta}{2\pi}$$

$$\theta = \frac{1}{2}\alpha t^2$$

$$\alpha = \frac{\omega}{\mathrm{t}} = \frac{60\pi}{60} = \pi$$

$\theta = \frac{1}{2} \times \pi \times 60 \times 60 = 1800\Pi$

$\mathrm{n} = \frac{\theta}{2\pi} = \frac{1800\pi}{2\pi} = 900$

13. $\omega = 200$ J,

$\Delta\vartheta = 9 - 4 = 5$ Hz

$\omega = 2\pi\vartheta$

$= 2\pi 5$

$= 10\pi$

$\mathrm{W} = \frac{1}{2}\mathrm{w}^2$

$200 = \frac{1}{2} I(10\pi)^2$

$\mathrm{I} = 0.15\mathrm{kgm}^2$

14. $\mathrm{L} = 90\mathrm{kgm}^2 5/$

$\Delta\omega = 40 - 20 = 20\mathrm{rad/s}$

$\mathrm{k} = \frac{1}{2}\mathrm{t}\omega$

$K = \frac{1}{2}(90)(20)$

$\mathrm{K} = 900$ J

15. $\mathrm{I} = 8 \times 10^{-2}\mathrm{kgm}^2$

$\omega = 20\mathrm{rad/s}$

$\tau = \mathrm{I}\alpha$

$\tau = \mathrm{I}\left(\frac{\mathrm{w}_2 - \mathrm{w}_1}{\mathrm{t}}\right)$

$\tau = 8 \times 10^{-2}\left(\frac{0 - 20}{4}\right)$

$\tau = 0.4\mathrm{Nm}$

16. $p = \tau\omega$

$100 \times 10 = \tau\left(\frac{1800 \times 2\pi}{60}\right)$

$\tau = 531$

17. $\tau = 5\mathrm{Nm}$

$\omega = \frac{420}{66} \times 2\pi\mathrm{rad/s}$

$p = \tau\omega$

$\mathrm{p} = (5)(14\pi)$

$\mathrm{p} = 220\mathrm{waff}$

18. $p = \tau\omega$

$6.28 \times 10^3 = \tau\left(\frac{3000}{60}\right)$

$\tau = \frac{6.28 \times 10^3}{50}$

$\tau = 20\mathrm{Nm}$

LEVEL – III

19. a-r, b-s, c-p, d-q

20. A, C, D

$\tau = 200\text{Nm}$
$\text{I} = 100\text{kgm}^2$
$\tau = \text{I}\alpha$
$2000 = 100\alpha$
$\alpha = 20.\,\text{rad/s}^2$
$\omega_1 = 0_1\ \alpha = 20\text{rad/s} = 10\text{sec}$
$\omega_2 = \text{w}_1 + \alpha\text{t}$
$\omega_2 = 0 + (20)(10)$
$\omega_2 = 200\text{rad/s}$

22. $\text{m} = 2\ \text{kg}$
$\text{r} = 5\ \text{m}$
$\alpha = 4\text{rad/s}^2$
$\tau = \text{I}\alpha$
$= \frac{2}{5}\text{mr}^2\alpha$
$= \frac{2}{5}(2)(5)^2(4)$
$\tau = 80\text{Nm}$

23. $\Delta\text{L} = 100\text{kgm}^2 5^{-1}$
$\Delta\omega = (50 - 40) = 10\text{rad/s}$
$1\text{k} = \frac{1}{2}\Delta\text{L}\Delta\omega$
$= \frac{1}{2}(100)(10) = \frac{1000}{5}\ \text{J}$
$\Delta\text{k} = 500\ \text{J}$

24. $\text{work} = \frac{1}{2}\text{MR}^2(\text{w}_2^2 - \text{w}_1^2)$
$= \frac{1}{2}(2)(25 \times 10^{-2})^2\left(\frac{20 \times 2\pi}{6}\right)^2$
$\omega = 0.2054\ \text{J}$

25. $\text{mk}^2 = \text{mr}^2$
$K = \sqrt{\frac{2}{5}R}$
$\text{I} = \frac{2}{5}\text{MR}^2$
$\tau = \text{I}\ \alpha$
$= \frac{2}{5}\text{MR}^2\alpha$

26. $\text{d} = \frac{\text{m}}{2\text{l}} \Rightarrow \text{M} = \text{d2I}$
$I = \frac{ML^2}{12} = \frac{d(21)(21)^2}{2^{12}}$
$= \frac{\text{d8I}^3}{12} = \frac{2\text{dl}^3}{}$

$$\rightarrow K = \frac{1}{2} I\omega^2$$

$$K = \frac{1}{2}\left(\frac{2dl^3}{3}\right)(\omega^2)$$

$$k = \frac{dl^3\omega^2}{3}$$

27. $\tau = I\alpha \Rightarrow \alpha = \frac{\tau}{I} = 0.3 \text{rad/s}^2$

$$\omega_2^2 - \omega_1^2 = 2\alpha\theta$$

$$\theta = \frac{\omega_2^2 - \omega_1^2}{2\alpha} \approx 7.855 \text{rev}$$